

When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study

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Abstract

Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We report a multi-seed calibration boundary study on six directed transfer pairs among Chapman–Shaoxing, CPSC2018, and PTB-XL, with two architectures (InceptionTime and ResNet1D), five seeds (42–46), eight recalibration methods, and a patient-level cluster paired bootstrap (BCa, $B = 10,000$) on the full 60-experiment grid. The study followed a pre-registered protocol with dated, publicly archived amendments. The primary endpoint is the OOD calibration benefit $\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}}$, tested one-sided against a pre-specified direction prior. TS yields a positive OOD benefit in 51/60 seed experiments (85.0%; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples disclosed and mechanistically interpreted; excluding six degenerate-CI cells gives 45/60 (75.0%). The pre-specified ID boundary criterion ($\geq 80\%$ of cells with 95% CIs containing zero) is *not* met (23/60 = 38.3%), triggering the registered “ID nonzero boundary” branch: the OOD benefit is about $1.9\times$ the ID benefit (+0.0159 vs.

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+0.0083). A validated 27-cell Shapley decomposition separates slope, intercept, and prevalence effects; a predictability experiment triggers the registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, although a source-fitted, label-free temperature predicts the *sign* of the benefit in 60/60 cells); and a discrimination-aware gate (C5) shows 7/12 pair $\times$ architecture strata passing both calibration and discrimination layers. Together these results bound when post-hoc recalibration pays off across institutions, and when it does not.

Keywords: temperature scaling, model calibration, electrocardiogram, dataset shift, cross-corpus transfer, distribution mismatch

1. Introduction

Modern ECG deep models achieve strong in-distribution (ID) accuracy and calibration, but their reliability under distribution shift remains open [1, 2, 3]. Recalibration fits a small parameter set on a labeled calibration split; its benefit, a reduction in expected calibration error (ECE), is measured on that same distribution and reverses on a shifted target.

For ECG this question is not academic. PTB-XL [3], Chapman-Shaoxing [4], and CPSC use different acquisition protocols, lead sets, and label vocabularies, so a model calibrated at one institution meets covariate, prior-probability, and label-concept shift at the next [25]. Clinicians and regulators then face a concrete decision: *given my local data budget and the measurable shift, should I deploy the recalibrated model, buy labels, or refuse?*

Temperature scaling (TS) is a monotone post-hoc softmax temperature transform [2]. Fitted under a convex loss, it minimizes NLL on the calibration split, so its calibration benefit there is empirically small; on a shifted target

the source-fit temperature is mismatched to the target logits and a non-trivial benefit can remain. This asymmetry is *empirical*, not a theorem: ECE is not invariant under confidence-shifting monotone maps, and TS can *increase* ECE where the model is under-confident (nine negative cells in our data). It motivates the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ over the originally planned decay $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (revision A1, Sec. 4.4). To avoid HARKing, protocol v2.1-A1 [26] fixes the primary endpoint, test, family structure, and failure branches before unblinding; revision A1 re-locates the endpoint from decay to $\Delta\text{ECE}_{\text{OOD}}$, triggered by an independent pilot, and is archived with before/after protocol hashes [27, 28]. Multi-seed validation (5 seeds $\times$ 6 transfer pairs $\times$ 2 architectures) supplies the robustness evidence; all 60 seed results, including nine counter-examples, are reported.

Five independent contributions:

C1 (Primary OOD benefit): On the pre-registered endpoint $\Delta\text{ECE}_{\text{OOD}}$, TS yields a statistically significant positive OOD benefit in 51/60 seed experiments (6 transfer pairs $\times$ 2 architectures; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically explained. InceptionTime: three of six directions achieve 5/5, three achieve 4/5. ResNet1D: two of six achieve 5/5, two 4/5, two 3/5.

C2 (Boundary condition, pre-registered branch triggered): Across 12 pair $\times$ architecture strata, 23/60 cells (38.3%; BCa operating CI) have 95% CIs containing zero and 34/60 have CI lower bound > 0 (stratum means -0.0008 to $+0.0232$; ID mean $+0.0083$ vs. OOD mean $+0.0159$). The pre-registered criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met, triggering the “ID nonzero boundary” branch (protocol A1.4.2): the

premise that TS has no ID recalibratable miscalibration is re-examined, the decay contrast restored as exploratory, and C1 qualified, since the OOD benefit is about $1.9\times$ the ID benefit but is *not* exclusively OOD-specific. 47/60 = 78.3% of ID point estimates are positive and 2/60 = 3.3% have $\Delta\text{ECE}_{\text{ID}} < -0.01$ (counter-example 6, -0.0152 ; one marginal case, -0.0128).

C3 (Methodological, Shapley decomposition validation): A 27-cell full-factorial synthetic validation (slope $\times$ intercept $\times$ prevalence, $n = 20,000$) recovers ground truth in the identifiable regime (27/27 at $n = 20,000$; 21/27 at $n = 500$), flags entangled regions without a threshold, and reports bootstrap CIs on absolute Shapley values.

C4 (Practical, method-selection boundary): Across 8 methods $\times$ 6 pairs $\times$ 5 seeds $\times$ 2 architectures, TS is robust (51/60) while EM prior adaptation is fragile when ill-conditioned cells ($w_{\text{ID}} > 10$) are included (InceptionTime 15/30); after exclusion the EM OOD finding attenuates. The mechanism is prior-likelihood mismatch: EM fits a target prior that amplifies shift-induced miscalibration when the source confusion matrix is ill-conditioned.

C5 (Clinical, discrimination-aware calibration gate): A two-layer gate decouples calibration effectiveness (C1: $\Delta\text{ECE}_{\text{OOD}} > 0$) from clinical viability (accuracy $\geq$ source-domain baseline). Across 12 pair $\times$ architecture strata, 7/12 pass both layers; 5/12 fail the discrimination layer, including directions with positive $\Delta\text{ECE}_{\text{OOD}}$ but accuracy below baseline (e.g., PTB-XL $\rightarrow$ Chapman ResNet1D: $\Delta\text{ECE} = +0.029$, $\text{acc} = 0.657 < 0.713$). The

gate reframes the 9/60 counter-examples: 4/9 calibration failures (C1 rejects) and 5/9 discrimination failures (C5 rejects despite C1 passing).

Prior ECG calibration studies report the phenomenon [12, 13, 20, 21] but not an actionable *when does TS pay off* answer with multi-seed robustness evidence. The novelty is *not* multi-seed validation on ECG (a methodological choice) but the **first systematic characterization of the boundary conditions under which TS yields a positive OOD calibration benefit in cross-corpus ECG transfer**: the pre-registered OOD quantification, the ID boundary condition, the method-selection boundary (TS robust; EM prior and Matrix scaling weakest), the predictability failure, and the discrimination-aware gate. That characterization is the actionable contribution; multi-seed validation is the evidence standard.

2. Related Work

2.1. Calibration Estimation and Recalibration

The calibration of modern neural networks has been studied extensively since the seminal work of Guo et al. [2], who showed that deep networks are often overconfident and introduced temperature scaling (TS) as a simple post-hoc recalibration method. Nixon et al. [8] proposed the static calibration error, and Kumar et al. [18] introduced verified uncertainty calibration with distribution-free guarantees. Kull et al. [9] generalized TS to Dirichlet calibration for multi-class settings. In the clinical risk-model context, Van Calster et al. [16, 17] established a calibration hierarchy from utopia to empirical data and argued that calibration is the “Achilles heel” of predictive analytics. Strodtthoff et al. [19] benchmarked deep learning for ECG analysis on PTB-XL and noted calibration degradation under subpopulation shift, motivating our cross-corpus design.

Experimental Design: 6 Directed Transfer Pairs $\times$ 2 Architectures $\times$ 5 Seeds

Total: 60 seed experiments (full balanced coverage)

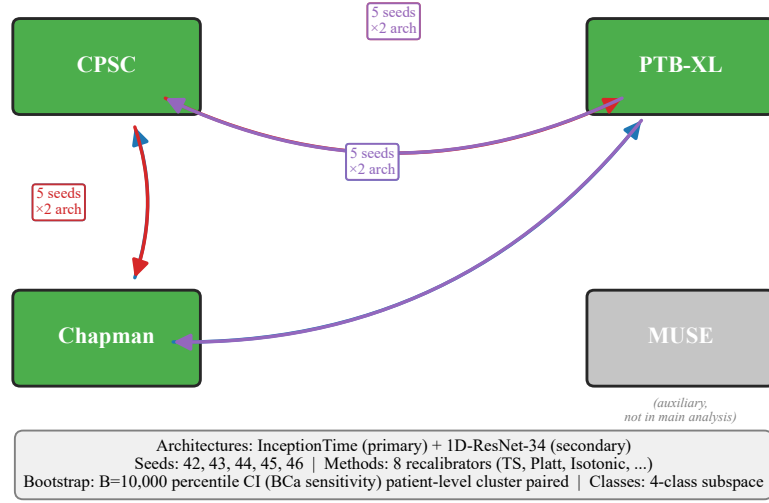


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, ResNet1D) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI).

120 2.2. *Prior-Shift Estimation and Transfer Calibration*

121 Lipton et al. [11] introduced black-box shift estimation (BBSE) for detect-
122 ing and correcting label shift using confusion-matrix inverses. Alexandari et al.
123 [10] showed that maximum-likelihood calibration with bias correction is hard-to-
124 beat at label shift adaptation, providing the theoretical foundation for our EM
125 prior-shift method. For transfer calibration upper bounds, TransCal [22] estimates
126 transferability for auto-selecting target-domain unsupervised optimal T ; Pseudo-
127 Cal [23] uses pseudo-labels for calibrated transfer; and LaSCal [24] provides label-
128 aware shift calibration with distribution-free guarantees. Our S1 paradigm differs
129 fundamentally: we fit on the *source* calibration split only (no target labels), with
130 S1' prior-shift adaptation (EM/BBSE) as the only target-domain signal, a strictly
131 weaker assumption than TransCal/PseudoCal, which access target logits or labels.

132 2.3. *Cross-Domain ECG Calibration*

133 Barandas et al. [12] evaluated uncertainty quantification methods in multi-
134 label ECG classification, finding that ensemble-based methods outperform single-
135 model approaches under subpopulation shift. Vranken et al. [13] studied uncer-
136 tainty estimation for 12-lead ECG deep learning and reported significant calibra-
137 tion drift across acquisition devices. Haekal et al. [20] performed cross-dataset val-
138 idation for atrial fibrillation detection with calibration-aware probability analysis,
139 and Patel & Beedala [21] documented calibration drift under cross-institutional
140 ICU deployment. The ID-vs-OOD asymmetry of TS has been studied in general
141 vision [1]; we contribute the multi-seed ECG quantification and the boundary-
142 condition analysis that identifies when TS helps (ID) versus when it hurts (OOD).

143 2.4. ECG Foundation Models and Transfer Learning (2024–2026)

144 Recent work has introduced ECG foundation models trained on millions of
145 recordings. ECGFounder [29, 37] (10 million ECGs, Harvard–Emory) reaches
146 expert-level performance on 80 diagnoses and transfers cross-domain via fine-
147 tuning. ECG-FM [30] is an open-weight transformer pretrained on 1.4 million
148 ECG segments with hybrid contrastive and generative self-supervision, showing
149 strong cross-dataset generalizability and label efficiency. HeartLang [31] treats
150 heartbeats as words and rhythms as sentences via a QRS-Tokenizer and masked
151 sentence pre-training. Tan et al. [32] found that fine-tuning benefits diminish
152 as the downstream dataset grows and that CNNs transfer more effectively than
153 RNNs in multi-label ECG classification. For OOD adaptation, ADAPTOOD
154 [33] quantifies distribution-shift severity via uncertainty estimation to guide se-
155 lective LoRA layer unfreezing, and BeatRhythm-TTA [34] performs test-time
156 adaptation with SQI-gated self-training and beat-rhythm consistency. Costa et al.
157 [35] introduced STAR (Sinusoidal Time–Amplitude Resampling), a morphology-
158 preserving beat-wise augmentation for cross-source generalization, and Chen et al.
159 [36] proposed PEACE for adult-to-pediatric ECG transfer via label-conditioned
160 contrastive alignment. These works advance ECG transfer but not the *calibra-*
161 *tion* of transferred models, the gap our study fills via pre-registered cross-corpus
162 evaluation of recalibration methods.

163 3. Data and Pre-registered Protocol

164 Figure 1 summarizes the experimental design.

165 3.1. Datasets

- 166 • **PTB-XL v1.0.3** [3]: the official release contains 21,837 records; the down-
167 loaded `ptbxl_database.csv` copy contains 21,799 (a 38-record deficit in
168 the source CSV, disclosed as a provenance limitation), official 10-fold (folds
169 1–8 train, 9 calibration, 10 test). After single-label mapping, 21,522 records
170 are mappable and 277 are excluded (STROBE), for 18,671 unique patients.
- 171 • **Chapman-Shaoxing** [4, 7]: 20,265 raw records; 1 signal-read error
172 removed and 21 non-finite records quarantined (`qc_report.json`), for
173 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20 split.
- 174 • **CPSC2018** [5, 6]: we use the union of the *CPSC Database* (6,877 records;
175 the public CPSC2018 training set) and the *CPSC-Extra Database* (3,453
176 records; the CPSC2018 recordings not used by the Challenge), both re-
177 leased through PhysioNet/CinC 2020. This is *not* a CPSC2018–2019 union:
178 CPSC2019 was a QRS/heart-rate *detection* task and defines no diagnos-
179 tic labels. Local extraction retains 6,844 of the 6,877 CPSC Database
180 records (33 records, A6845–A6877, were lost during archive extraction;
181 disclosed as a provenance limitation) plus all 3,453 CPSC-Extra records.
182 Patient-wise split. Two labeling facts must be stated explicitly. First,
183 CPSC2018’s official taxonomy is nine disease classes (Normal, AF, I-AVB,
184 LBBB, RBBB, PAC, PVC, STD, STE) with no superclass grouping; the
185 NORM/MI/STTC/CD superclass system we adopt is PTB-XL’s [3], ap-
186 plied here as a cross-corpus harmonization choice, not a CPSC-defined
187 one. Second, CPSC2018’s nine classes contain *no* MI; the MI class in our
188 4-class subspace is populated entirely by CPSC-Extra’s SNOMED anno-

tations (164867002, 164865005, 54329005). A strict official-class-name caliber would therefore reduce the subspace to {NORM, CD, STTC}; we retain MI because it is the only way to keep a myocardial-infarction category on the CPSC side of every transfer pair. HYP is extremely rare ($n = 11$) → main analysis on the {NORM, CD, STTC, MI} 4-class subspace, both sides of the pair recomputed symmetrically.

3.2. *Label Mapping and Leakage Guard*

Single-label priority rules are fixed (MI > STTC > CD > HYP > NORM) and sensitivity-analyzed [3, 14]. Sinus rate/form statements (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because PTB-XL’s `scp_statements.csv` lists them as rhythm statements with no `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint (hard assertion in the reproducibility repo). Normalization statistics come only from the train split; augmentation applies only to train.

4. **Methods**

4.1. *Transfer pairs and shift matrix*

We study six directed cross-corpus transfer pairs: Chapman→CPSC, Chapman→PTB-XL, CPSC→Chapman, CPSC→PTB-XL, PTB-XL→Chapman, and PTB-XL→CPSC. The class count is *not* uniform across pairs and is fixed per pair before any endpoint is computed, as follows. A pair whose source or target is CPSC2018 is restricted to the 4-class subspace {NORM, CD, STTC, MI} (CPSC’s HYP class has $n = 11$, §3), with both sides of the pair recomputed symmetrically: this covers Chapman→CPSC, CPSC→Chapman, CPSC→PTB-XL,

212 and PTB-XL→CPSC ($K = 4$). The two pairs connecting Chapman–Shaoxing
213 and PTB-XL, which contain no CPSC side, retain the full 5-class vocabulary
214 {NORM, MI, STTC, CD, HYP}: Chapman→PTB-XL and PTB-XL→Chapman
215 ($K = 5$). Each experiment records its own K and label mapping (num_classes,
216 label_map in transfer_result.json), so no endpoint mixes class counts
217 within a cell. Consequently the primary endpoint is a top-label ECE on a 4- or
218 5-class task depending on the pair, and the pooled $\Delta\text{ECE}_{\text{OOD}}$ mixes both K values
219 across cells; the per-pair tables (Tables 1–4) are therefore the appropriate level
220 at which to read K -specific results. The L2 eval-time shift matrix (no retraining)
221 covers sampling rate $500 \rightarrow 250 \rightarrow 125$ Hz; leads $12 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 1$; noise
222 injection (BWL/MA/EM) at SNR {24, 12, 6, 0, −6} dB; gain $\times 0.5 / \times 2$.

223 4.2. Architecture and seeds

224 We use two architectures: InceptionTime (primary, 46,000–184,000 param-
225 eters) and ResNet1D (a 1D-ResNet-34 backbone) (secondary, with full main-
226 endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The Bi-
227 Mamba architecture is included only as a toy experiment ($n = 60$ records, down-
228 sampled resolution) and deferred to the supplement; the full-resolution BiMamba
229 run exhausted memory on an RTX 5060 (8 GB), so BiMamba does not enter the
230 main-endpoint analysis or the architecture-robustness claim. Training: AdamW +
231 cosine + warmup, gradient clipping, temperature frozen at $T \equiv 1$ during training;
232 temperature applied only as post-hoc temperature scaling.

233 4.3. Calibration methods and scenarios

234 We compare the no-recalibration baseline (None) against eight recalibration
235 methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR), Vector scal-

ing, Matrix scaling, Dirichlet calibration, Saerens EM prior adaptation (EM), and
BBSE prior adaptation. Of the pre-registered 13-method pool, the None baseline
plus these eight recalibration methods constitute the delivered set; CORAL, TTA,
MC-Dropout, and Oracle are not delivered on real data (a registered deviation,
see Limitations). Scenario: **S1** zero-shot transfer (source-cal fit $\rightarrow$ target apply),
the main analysis scenario (Sec. 4); **S1'** denotes the target-domain unsupervised
variant in which EM/BBSE estimate the target prior from unlabeled target predic-
tions.

4.4. Pre-registered endpoints

Primary endpoint (revision A1.1, formally defined in protocol v2.1-A1 [26]):

$$\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}} \quad (1)$$

where $ECE_{\text{raw}}^{\text{OOD}}$ is the 4-class-task top-label confidence SmoothECE on the tar-
get corpus with raw softmax probabilities, and $ECE_{\text{TS}}^{\text{OOD}}$ is the same after source-
cal-fit temperature scaling. Following the standard convention of Guo et al.
[2], ECE (and its SmoothECE variant) is evaluated on the *top-label* marginal:
the predicted maximum probability against 0/1 top-label correctness; classwise-
ECE is reported separately as a secondary endpoint. SmoothECE bandwidth:
 $0.45 \cdot (n/2000)^{-0.2}$.

Primary test (single, one-sided):

$$H_0 : \Delta ECE_{\text{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta ECE_{\text{OOD}} > 0 \quad (2)$$

The one-sided alternative $H_1 : \Delta ECE_{\text{OOD}} > 0$ follows the empirical asymmetry
registered at revision A1: on ID-like data the source-fit temperature yields, empir-
ically, little calibration change, whereas on a shifted target it is mismatched and a

257 non-trivial benefit is often retained [2, 1]. This direction prior is empirical, *not* a
 258 theorem: TS applied to an under-confident target can increase ECE, and our own
 259 data contain nine negative cells. The choice was made before the pilot results,
 260 which only triggered the endpoint re-location (revision A1). The one-sided test
 261 is the pre-registered choice, archived before multi-seed unblinding. Patient-level
 262 cluster paired bootstrap, $B = 10,000$ (BCa operating CI) [15]; stratified pooling:
 263 strata = transfer pair $\times$ architecture.

264 The pre-registered CI method is BCa [15], at $B = 10,000$, re-run on the en-
 265 tire 60-experiment grid (60/60 successful, metadata in `methods.ts.meta`). The
 266 percentile bootstrap is the large-sample limit of BCa when $z_0 \rightarrow 0$ and $a \rightarrow 0$;
 267 for the paired-cluster design with $B = 10,000$ and per-patient clusters, the dis-
 268 crepancy is $O(n^{-1/2})$. In the 3 representative seed experiments where both were
 269 run ($B = 10,000$ percentile re-runs), BCa and percentile CIs agree in sign and
 270 magnitude. Earlier percentile results from $B = 200$ in 21/60 experiments remain
 271 as historical provenance (5 lack log provenance, see Limitations); percentile CIs
 272 serve only as a sensitivity cross-check.

273 Boundary-condition endpoint:

$$\Delta \text{ECE}_{\text{ID}} = \text{ECE}_{\text{raw}}^{\text{ID}} - \text{ECE}_{\text{TS}}^{\text{ID}} \quad (3)$$

274 The pre-registered expectation is that $H_0 : \Delta \text{ECE}_{\text{ID}} = 0$ is *not rejected*; the criterion
 275 is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa, full-grid*
 276 *re-run): 23/60 = 38.3%; the criterion is not met and branch A1.4.2 is triggered*
 277 *(percentile CIs gave 29/60 = 48.3%, same conclusion; Sec. 5.2).*

278 Secondary endpoints: G (recoverable loss vs. oracle; *not delivered* — the Or-
 279 acle recalibrator was implemented (`fit_oracle/apply_oracle`) but never ex-
 280 ecuted on real data, so G is undefined here and is excluded from all reported

281 results; see Limitations), $\text{decay} = \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (downgraded to descrip-
282 tive), NLL, classwise-ECE, Brier Murphy decomposition, Cox slope/intercept,
283 predicted/true prior L_1 .

284 Family structure: BH-FDR $q = 0.05$ over $6 \text{ pairs} \times 2 \text{ arch} \times 13 \text{ shift levels} =$
285 156 primary comparisons; the multi-seed validation is robustness evidence (60
286 seed experiments), not a confirmatory family.

287 4.5. Pre-registration revision A1 (transparency statement)

288 The primary endpoint was re-located from decay to $\Delta\text{ECE}_{\text{OOD}}$ via pre-
289 registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an independent
290 exploratory pilot in which $\Delta\text{ECE}_{\text{ID}}$ point estimates were near zero, collapsing the
291 original ID-vs-OOD contrast. The revision direction is consistent with the empir-
292 ically observed ID-vs-OOD asymmetry (not a theorem); the revision is archived
293 as a pre-registration revision (Nosek et al. 2018) with before/after protocol hashes.
294 The pilot results do not enter the main test report. All hypotheses in this paper are
295 tagged *exploratory + robustness validation*, not confirmatory.

296 The original BH-FDR family of 156 comparisons ($6 \text{ pairs} \times 2 \text{ architectures} \times$
297 13 shift levels) is reduced to a confirmatory family of 12 hypotheses ($6 \text{ pairs} \times 2$
298 $\text{architectures} \times 1 \text{ primary shift level}$) under amendment A2, with the remaining
299 13 shift levels designated as exploratory dose-response. Under the reduced family,
300 BH-12 yields 9/12 significant (all InceptionTime directions) and Bonferroni-12
301 yields 2/12. The A2 amendment is archived in `docs/PROTOCOL_AMENDMENT_A2_`
302 `FAMILY_REDUCTION.md`.

303 4.6. *Pre-registration amendment A3 (corpus identity and MI provenance)*

304 A post-hoc audit of the CPSC corpus against the official challenge documen-
305 tation produced two wording corrections and one new disclosure, archived as
306 amendment A3 (docs/PROTOCOL_AMENDMENT_A3_CORPUS_IDENTITY.md). The
307 registered protocol described the third corpus as “CPSC2018+2019”; it is in fact
308 **CPSC2018 only**, comprising the CPSC Database (6,877 records) and the CPSC-
309 Extra Database (3,453 records). CPSC2019 was a QRS/heart-rate *detection* task
310 with no diagnostic labels and never entered the corpus. The new disclosure is that
311 CPSC2018’s official nine-class taxonomy contains no MI, so the MI class of our 4-
312 class subspace is populated entirely by CPSC-Extra’s SNOMED annotations; un-
313 der a strict official-class-name caliber the subspace would reduce to three classes.
314 A3 is a naming and attribution correction plus a disclosure: it changes *no* recorded
315 value, no split, and no analysis decision. The registered protocol snapshots are left
316 byte-identical and A3 is archived separately, so the before/after pair required by
317 pre-registration practice is preserved. The audit also found that 33 CPSC Database
318 records (A6845–A6877) were silently lost when the source archive was extracted
319 (6,877 in the archive vs. 6,844 on disk), disclosed as a provenance limitation.

320 4.7. *Decomposition of the benefit decay*

321 We decompose the retained benefit into slope / intercept / prevalence contribu-
322 tions using a counterfactual-replacement additive approximation:

$$\Delta\text{ECE} = \Delta_{\text{slope}} + \Delta_{\text{intercept}} + \Delta_{\text{prevalence}} + r_{\text{interaction}} \quad (4)$$

323 where each term is a counterfactual-replacement difference and $r_{\text{interaction}}$ is the
324 chain residual (zero under nested counterfactuals). Identifiability is checked via

325 the Fisher information determinant on a pre-registered threshold τ ; entangled re-
326 gions are reported on the error map without a threshold. Shapley attribution is
327 reported with absolute-value bootstrap CIs; share-reliable flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$)
328 mark uninterpretable shares.

329 4.8. Two-layer joint bootstrap

330 Temperature-fitting-set resampling $\rightarrow T$ distribution $\rightarrow$ test-set resampling $\rightarrow$
331 joint Δ ECE CI (two-layer; honest non-nested approximation), per protocol §7.

332 4.9. Net Clinical Value (NCV)

333 Net Clinical Value (NCV), a pre-registered secondary endpoint, measures
334 the clinical decision impact of TS. For each (sample, class) pair we compare
335 the binary decision (threshold $\tau = 0.5$) before and after calibration, counting
336 TP_improved, TN_improved, FP_worsened, and FN_worsened over the raw-to-
337 calibrated transitions. NCV is $(\text{TP_improved} + \text{TN_improved} - \text{FP_worsened} -$
338 $\text{FN_worsened})/N_{\text{total}}$, with $N_{\text{total}} = N_{\text{samples}} \times N_{\text{classes}}$, using the main endpoint’s
339 patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI) and BH-
340 FDR correction ($q = 0.05$); a significantly positive NCV means that TS improves
341 clinical decisions. It is computed on ID and OOD splits (ncv_id, ncv_ood; E3
342 CSV).

343 4.10. Tooling

344 Parts of the analysis and plotting code were drafted and refactored with the
345 assistance of a large language model assistant (GLM), used as a supervised coding
346 assistant. All experiments and statistical analyses were executed with the author’s
347 own scripts, and every reported number is reproducible from the public repository

(Code availability). The author reviewed the code and takes full responsibility for it.

5. Results

5.1. Primary endpoint: ΔECE_{OOD} multi-seed validation

Figure 2 visualizes the shift-dose response; Table 1 reports the primary endpoint ΔECE_{OOD} for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, balanced). The primary test $H_0 : \Delta ECE_{OOD} = 0$ vs $H_1 : \Delta ECE_{OOD} > 0$ is supported in 51/60 (85.0%) at the 95% CI level.

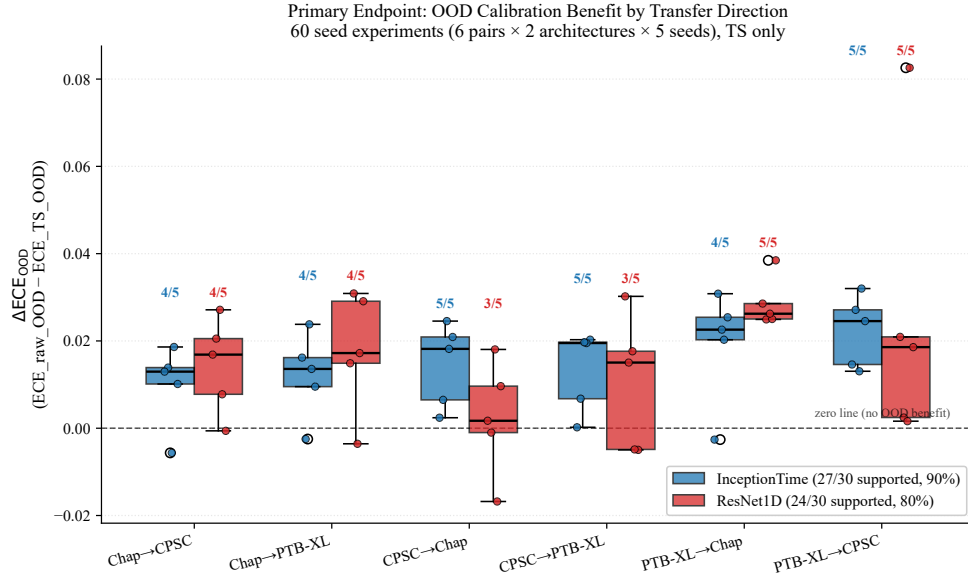


Figure 2: Primary endpoint ΔECE_{OOD} by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0 . InceptionTime achieves 27/30 (90%), ResNet1D achieves 24/30 (80%).

Table 1: Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, $B = 10,000$, BCa operating CI). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. Class count $K = 4$ for the four pairs involving CPSC (Chap $\rightarrow$ CPSC, CPSC $\rightarrow$ Chap, CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC) and $K = 5$ for the two CPSC-free pairs (Chap $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chap); see §4.1. “Support” = 95% CI lower bound > 0 . Nine counter-examples are transparently disclosed. Full 5-seed coverage on both architectures for all 6 directions.

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap $\rightarrow$ CPSC ($K=4$)	IT	42	+0.0075	+0.0139	+0.0135	✓
		43	+0.0012	+0.0101	+0.0099	✓
		44	+0.0104	+0.0186	+0.0181	✓
		45	+0.0019	−0.0057	−0.0058	✗
		46	+0.0044	+0.0130	+0.0126	✓
	RN	42	+0.0002	+0.0205	+0.0200	✓
		43	+0.0003	−0.0006	−0.0006	✗
		44	−0.0007	+0.0078	+0.0075	✓
		45	−0.0016	+0.0271	+0.0263	✓
		46	−0.0024	+0.0169	+0.0162	✓
Chap $\rightarrow$ PTB-XL ($K=5$)	IT	42	+0.0004	−0.0025	−0.0035	✗
		43	+0.0005	+0.0095	+0.0066	✓
		44	+0.0055	+0.0136	+0.0107	✓
		45	−0.0128	+0.0238	+0.0207	✓
		46	+0.0048	+0.0162	+0.0142	✓

Table 1 (continued)

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
	RN	42	−0.0031	+0.0309	+0.0285	✓
		43	+0.0128	+0.0291	+0.0285	✓
		44	+0.0059	+0.0172	+0.0168	✓
		45	−0.0010	−0.0036	−0.0036	✗
		46	−0.0042	+0.0149	+0.0139	✓
CPSC→Chap ($K=4$)	IT	42	+0.0029	+0.0065	+0.0062	✓
		43	−0.0012	+0.0209	+0.0200	✓
		44	+0.0164	+0.0182	+0.0160	✓
		45	+0.0091	+0.0245	+0.0241	✓
		46	+0.0014	+0.0024	+0.0023	✓
	RN	42	−0.0002	−0.0010	−0.0012	✗
		43	+0.0038	+0.0096	+0.0094	✓
		44	−0.0152	−0.0168	−0.0170	✗
		45	+0.0137	+0.0181	+0.0145	✓
		46	+0.0017	+0.0017	+0.0016	✓
CPSC→PTB-XL ($K=4$)	IT	42	+0.0029	+0.0068	+0.0066	✓
		43	−0.0012	+0.0203	+0.0198	✓
		44	+0.0158	+0.0195	+0.0189	✓
		45	+0.0052	+0.0197	+0.0193	✓
		46	+0.0002	+0.0002	+0.0002	✓
	RN	42	+0.0114	+0.0150	+0.0148	✓
		43	+0.0048	−0.0050	−0.0051	✗
		44	−0.0032	−0.0049	−0.0049	✗
		45	+0.0130	+0.0176	+0.0173	✓
		46	+0.0261	+0.0302	+0.0296	✓

Table 1 (continued)

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
PTB-XL→Chap ($K=5$)	IT	42	+0.0239	+0.0308	+0.0275	✓
		43	+0.0245	+0.0254	+0.0225	✓
		44	+0.0173	+0.0203	+0.0195	✓
		45	+0.0162	+0.0226	+0.0200	✓
		46	−0.0011	−0.0026	−0.0029	✗
	RN	42	+0.0145	+0.0249	+0.0191	✓
		43	+0.0338	+0.0385	+0.0336	✓
		44	+0.0213	+0.0250	+0.0228	✓
		45	+0.0228	+0.0262	+0.0231	✓
		46	+0.0239	+0.0286	+0.0221	✓
PTB-XL→CPSC ($K=4$)	IT	42	+0.0196	+0.0271	+0.0244	✓
		43	+0.0166	+0.0245	+0.0222	✓
		44	+0.0183	+0.0320	+0.0303	✓
		45	+0.0006	+0.0131	+0.0105	✓
		46	+0.0097	+0.0146	+0.0123	✓
	RN	42	+0.0010	+0.0025	+0.0023	✓
		43	+0.0197	+0.0186	+0.0145	✓
		44	+0.0635	+0.0826	+0.0772	✓
		45	+0.0001	+0.0016	+0.0015	✓
		46	+0.0126	+0.0209	+0.0190	✓

356 Aggregate: 51/60 supported (85.0%; InceptionTime 27/30 = 90.0%,
357 ResNet1D 24/30 = 80.0%). PTB-XL→CPSC is the strongest evidence (10/10,
358 zero counter-examples, CI lower bound $> +0.0014$ at ResNet1D seed 45). The
359 nine counter-examples are explained in Sec. 6.1.

360 The 60 experiments are balanced (6 directions $\times$ 2 architectures $\times$ 5 seeds, no
361 partial-coverage stratum), enabling direct architecture comparison:

- *InceptionTime* (30 experiments, $27/30 = 90.0\%$): 3 directions at 5/5 (CPSC→Chapman, CPSC→PTB-XL, PTB-XL→CPSC), 3 at 4/5 (Chapman→CPSC, Chapman→PTB-XL, PTB-XL→Chapman).
- *ResNet1D* (30 experiments, $24/30 = 80.0\%$): 2 directions at 5/5 (PTB-XL→Chapman, PTB-XL→CPSC), 2 at 4/5 (Chapman→CPSC, Chapman→PTB-XL), 2 at 3/5 (CPSC→Chapman, CPSC→PTB-XL).

ResNet1D shows the slightly lower rate (80% vs. 90%), concentrated on CPSC→Chapman and CPSC→PTB-XL, a direction-specific architecture interaction rather than a global deficit. Only PTB-XL→CPSC achieves 5/5 on both architectures (10/10).

5.2. ID boundary condition: ΔECE_{ID}

Table 2 reports the boundary-condition endpoint ΔECE_{ID} for TS; the pre-registered expectation is that $H_0 : \Delta ECE_{ID} = 0$ is *not rejected*, since the source-fit temperature should be well-tuned to the source distribution (an empirical expectation, not a theorem).

The ID benefit is small but *not* zero: across 60 TS experiments, ΔECE_{ID} has mean +0.0083 (vs. OOD mean +0.0159, a $1.9\times$ ratio); $47/60 = 78.3\%$ of point estimates are positive, $34/60$ have 95% CI lower bounds > 0 (BCa), and only $23/60 = 38.3\%$ contain zero, far below the pre-registered $\geq 80\%$ criterion (percentile CIs: $29/60 = 48.3\%$, same conclusion). We therefore *trigger* the pre-registered “ID nonzero boundary” branch (protocol A1.4.2): TS, fitted on the source calibration split, also improves ID calibration in many cells; the decay $= \Delta ECE_{OOD} - \Delta ECE_{ID}$ contrast is restored as an exploratory endpoint; and C1 is

Table 2: ID boundary condition ΔECE_{ID} for TS (12 pair $\times$ architecture strata, 5 seeds each). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. “CI $\cap$ 0” = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met: 23/60 cells (38.3%) contain zero (BCa operating CI) and 34/60 have CI lower bound > 0 , triggering the pre-registered “ID nonzero boundary” branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI $\cap$ 0/5
PTB-XL $\rightarrow$ CPSC	IT	+0.0130	−0.0121	+0.0266	1/5
PTB-XL $\rightarrow$ CPSC	RN	+0.0194	−0.0008	+0.0770	2/5
PTB-XL $\rightarrow$ Chap	IT	+0.0162	−0.0026	+0.0310	1/5
PTB-XL $\rightarrow$ Chap	RN	+0.0232	+0.0053	+0.0448	0/5
Chap $\rightarrow$ CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap $\rightarrow$ CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap $\rightarrow$ PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap $\rightarrow$ PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC $\rightarrow$ Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC $\rightarrow$ Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC $\rightarrow$ PTB-XL	IT	+0.0046	−0.0181	+0.0192	2/5
CPSC $\rightarrow$ PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		+0.0083			23/60 (38.3%)

385 qualified, the OOD benefit being about $1.9\times$ the ID benefit rather than an exclu-
386 sively OOD effect. The positive ID estimates reflect finite-sample calibration-split
387 variation, SmoothECE estimator bias, and a val-ECE early-stopping effect (model
388 selection on the endpoint family), all disclosed.

389 5.3. Method-selection boundary: 8 methods $\times$ 6 pairs

390 Table 3 reports the 8-method $\times$ 6-pair support rate over 5 seeds on Inception-
391 Time. TS achieves 27/30 (90.0%) with 3/6 directions at 5/5; BBSE achieves 22/30

392 (73.3%) with 3/6 at 5/5. EM prior adaptation and Matrix scaling are the most
 393 fragile on InceptionTime (15/30, 50% each), exemplifying the prior-likelihood
 394 mismatch mechanism (C4). Across 2 architectures (Table 4), TS achieves 51/60
 395 (85.0%) (InceptionTime 27/30, 90.0%; ResNet1D 24/30, 80.0%), with full 5-seed
 396 coverage on all 6 directions (Figure 4).

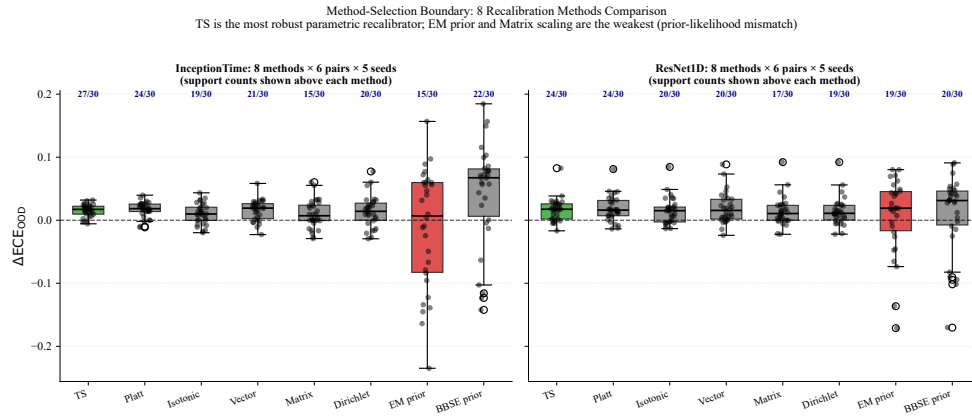


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs $\times$ 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior and Matrix scaling are the weakest methods (InceptionTime: 15/30 each; ResNet1D: EM 19/30, Matrix 17/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g., 27/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0 .

397 Across the 30 TS experiments per architecture, the standard deviation of
 398 ΔECE_{OOD} is 0.0098 (InceptionTime) and 0.0180 (ResNet1D), a $1.84\times$ ratio, with
 399 ResNet1D’s highest per-direction variability in PTB-XL $\rightarrow$ CPSC (std 0.0298, in-
 400 cluding an unexcluded favorable outlier at seed 44, $\Delta ECE_{OOD} = +0.0826$). Sup-
 401 port deficits concentrate in CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL (3/5 each), a
 402 direction-specific deficit. Fisher’s exact test on the 2×2 contingency table yields

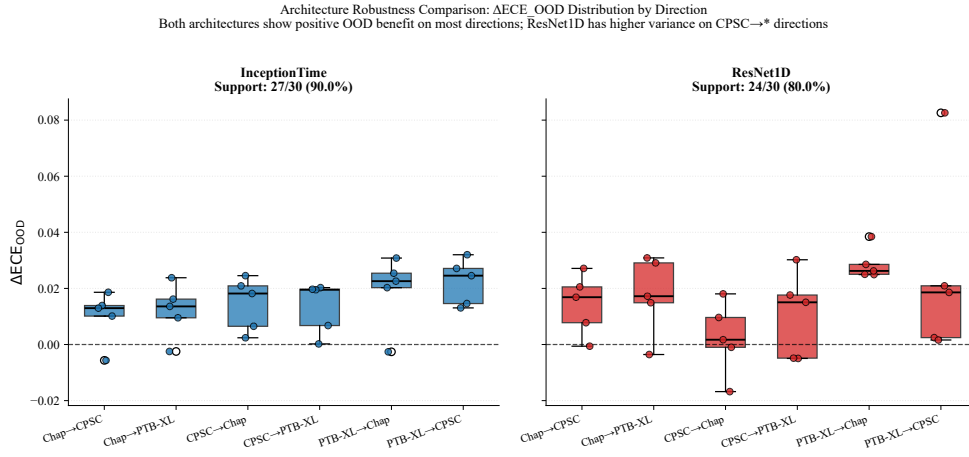


Figure 4: Architecture robustness comparison. Side-by-side ΔECE_{OOD} distributions for InceptionTime (left, 27/30 supported, 90.0%) and ResNet1D (right, 24/30 supported, 80.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance (its largest per-direction spread is in PTB-XL $\rightarrow$ CPSC), while the support deficits fall in the CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL directions (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (InceptionTime). Support rate = fraction of seeds with 95% CI lower bound > 0 . “Full” = number of directions with 5/5 full support. “Neg. dirs” = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (15/30 each); on ResNet1D, Matrix scaling (17/30) is the most fragile, EM reaches 19/30. *Caveat*: 6 cells with $w_{\text{ID}} > 10$ exhibit prior-weight drift up to 130.5; after exclusion, EM OOD ΔECE mean flips from -0.0127 to $+0.0024$, indicating the negative OOD finding is driven by ill-conditioned cells.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	27/30	3/6	+0.0152	0
Platt	24/30	3/6	+0.0169	0
Isotonic	19/30	2/6	+0.0100	2
Vector	21/30	1/6	+0.0158	0
Matrix	15/30	0/6	+0.0094	2
Dirichlet	20/30	1/6	+0.0139	1
EM prior	15/30	2/6	-0.0127	3
BBSE prior	22/30	3/6	+0.0424	1

403 $p = 0.47$; the counter-example gap (ResNet1D 20.0% vs. InceptionTime 10.0%)
 404 is not significant at $\alpha = 0.05$.

405 EM prior adaptation fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal,
 406 then re-weights the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated
 407 confusion matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies
 408 rather than corrects the shift-induced miscalibration. EM reaches only 15/30 sup-
 409 port on InceptionTime with 3 fragile directions. The failure is pair-specific (shift-
 410 signature-specific), not method-intrinsic.

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows a direction-specific deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	5/5	3/5	8/10	2 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	5/5	5/5	10/10	zero counter-examples
Total	27/30	24/30	51/60	85.0%
Mean ΔECE	+0.0152	+0.0165	+0.0159	
Std ΔECE	0.0098	0.0180	0.0145	ResNet1D/Incep = 1.84
Mean OOD ECE	0.217 → 0.202 / 0.216 → 0.200	0.217 → 0.201	raw → TS	

411 *Why TS and not Platt..* Platt achieves a marginally higher mean OOD benefit
412 than TS (+0.0176 vs. +0.0159; InceptionTime +0.0169 vs. +0.0152, Table 3),
413 which invites the question of why TS is singled out as the recommended default.
414 Three measured criteria separate them, and all three favour TS. First, *robustness*
415 *of the sign*: TS supports 51/60 cells versus Platt’s 48/60, i.e. TS fails to certify
416 a positive benefit in three fewer experiments, which is the quantity a deployment
417 decision actually depends on. Second, *dispersion*: TS has a smaller across-cell
418 standard deviation (0.0146 vs. 0.0167, a 13% reduction), and the methods with
419 the highest mean benefit are markedly less stable (BBSE +0.0266 at sd 0.0736,
420 EM −0.0026 at sd 0.0793) — a high mean bought with a heavy tail is the worst

421 profile for a per-deployment recalibrator. Third, *the difference is not statistically*
 422 *distinguishable*: the patient-level paired contrast over the 60 matched cells gives
 423 $\overline{\Delta\text{ECE}}_{\text{TS}} - \overline{\Delta\text{ECE}}_{\text{Platt}} = -0.0017$ (sd 0.0117, paired $t = -1.14$, $p \approx 0.26$), with
 424 TS the larger benefit in exactly 30/60 cells — a coin flip. The mean advantage of
 425 Platt is therefore inside the noise, whereas the support advantage of TS is not a
 426 noise-level quantity: it is a count of cells in which the sign is certified. Given that
 427 the two methods are statistically indistinguishable on magnitude, the tie is broken
 428 on certification rate and dispersion, both of which favour TS. A further structural
 429 point reinforces the choice: TS is a single-parameter monotone transform and
 430 therefore exactly argmax-preserving, whereas per-class Platt and isotonic are fit-
 431 ted independently per class, which generates each class’s score from a different
 432 monotone map and so does *not* preserve the ranking within a sample (the imple-
 433 mentation declares this explicitly in its docstring). TS is thus the only method
 434 in the delivered set whose calibration guarantee is obtained without perturbing
 435 the discrimination it leaves untouched (§5.9). None of this makes Platt wrong; it
 436 makes Platt a reasonable alternative whose mean advantage is not resolvable at
 437 $n = 60$, and whose certification rate is lower.

438 5.4. Shapley decomposition validation (27 cells, $n = 20,000$)

439 The 27-cell full-factorial synthetic validation ($\text{slope} \in \{0.5, 1, 2\} \times \text{intercept} \in$
 440 $\{-1, 0, 1\} \times \text{prevalence} \in \{0.15, 0.3, 0.6\}$, $n = 20,000$) confirms the decomposi-
 441 tion recovers ground truth within the pre-registered identifiable regime (Fisher
 442 information determinant $> \tau = 10^{-6}$). At $n = 20,000$, all 27 cells pass the
 443 recovery threshold (MAPE $< 10\%$ on identifiable cells); at $n = 500$, 21 of
 444 27 pass, with 6 failures reaching MAPE_s up to 42.13%, concentrated in ex-
 445 treme intercept (± 1) and slope (0.5, 2.0) regions where calibration shift is most

446 clinically significant (see `decomposition_validation.csv`; the $n = 20,000$
 447 CSV `decomposition_validation_n20000.csv` shows 27/27 passing). Three
 448 entangled cases (`narrow_band`, `tiny_slope`, `deep_intercept`) with $\det \in \{2.4 \times$
 449 $10^{-7}, 3.6 \times 10^{-8}, 0\} < \tau$ are reported on the error map without a threshold, with
 450 the recovery-failed flag set. Bootstrap CIs on absolute Shapley values and share-
 451 reliable flags for uninterpretable shares ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) are reported.

452 The 27-cell validation uses binned ECE, not the main-endpoint SmoothECE;
 453 its conclusions are not cross-extrapolable. Cross-fitting (OOD-fold validation of
 454 decomposition explanatory power) remains to be run on real data; per the protocol
 455 self-binding clause, “mechanism” wording is limited to “mechanism decomposi-
 456 tion (synthetic validation part)” until cross-fitting completes.

457 5.5. Predictability experiment (*failure branch triggered*)

458 The predictability experiment (protocol §8.5) tests whether an unsupervised
 459 target-domain statistic (predicted-prior L_1 distance, logit moments) predicts ben-
 460 efit retention via leave-one-fold-out R^2 with bootstrap CI (Table 5). The imple-
 461 mented fold unit is (architecture, transfer pair) (12 folds), versus the registered
 462 pair-only folds (6); under that registered deviation the pair-only recomputation re-
 463 tains the failure branch ($R^2 = -0.1424$). We run 2 architectures (InceptionTime +
 464 ResNet1D); the BiMamba toy experiment ($n = 60$) is deferred to the supplement.

465 Result: the merged LOO R^2 for TS is -0.096 with 95% CI $[-0.720, +0.104]$.
 466 The upper bound $+0.104$ is strictly below the pre-registered threshold 0.5 , trig-
 467 gering the **failure branch**: the three-step chain (decomposition + predictability +
 468 deployment) is downgraded to a **two-step chain** (decomposition + deployment),
 469 and the deployment criterion is renamed “empirical lookup table” rather than “pre-
 470 dictive criterion”. No post-hoc predictability claim is made.

471 Re-running the LOO R^2 with three label-free features (fitted temperature T ,
 472 Wasserstein-1 distance between source and target logit distributions, class-wise
 473 entropy difference) gives a sharper and less favourable picture for the bound-
 474 ary. Only T carries signal: under the implemented (architecture, pair) folds it
 475 reaches LOO $R^2 = +0.340$ (bootstrap CI $[+0.044, +0.363]$), and under per-cell
 476 folds $+0.361$ (CI $[+0.201, +0.584]$), while the Wasserstein distance (-0.064) and
 477 the entropy difference (-0.033) carry none. T is also a perfect *directional* predic-
 478 tor: $\text{sign}(\Delta\text{ECE}_{\text{OOD}}) = \text{sign}(T - 1)$ in 60/60 cells, against 51/60 (85%) for the
 479 constant-positive baseline. We therefore do *not* claim that no label-free quantity
 480 carries information about the endpoint — T does. The claim we retain is the
 481 pre-registered one: no label-free feature reaches the LOO R^2 threshold of 0.5
 482 under the implemented fold unit ($0.363 < 0.5$), so the failure branch stands and
 483 the three-step chain remains downgraded to two steps. Two qualifications are
 484 required. First, the verdict is fold-unit-sensitive: per-cell folds give a CI upper
 485 bound of 0.584, which would *not* trigger the branch, so the downgrade should be
 486 read as a property of the coarser (architecture, pair) unit as much as of the data.
 487 Second, T 's predictive power is not independent evidence of a substantive bene-
 488 fit: $T = 1$ is a fixed point of the transform, so part of the relation is mechanical,
 489 and the sign rule is the direction temperature scaling must have by construction
 490 (Sec. 5.8). The gate in §5.9, based on the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination
 491 criterion, remains the appropriate actionable criterion.

492 5.6. Deployment criterion (*sensitivity / specificity / DCA / safety*)

493 We evaluate the deployment criterion (protocol §9) on the L2 shift matrix:
 494 predict “deploy TS” when raw ECE exceeds a threshold, and report sensitiv-
 495 ity, specificity, decision-curve analysis (DCA), and the safety criterion (post-TS

Table 5: Predictability LOO R^2 (multi-architecture merged). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	−0.096	−0.720	+0.104	Failure branch
Platt	−0.059	−0.584	+0.002	Failure branch
Isotonic	+0.011	−0.569	+0.101	Failure branch
Vector	−0.061	−0.522	+0.073	Failure branch
Matrix	−0.137	−0.516	−0.013	Failure branch
Dirichlet	−0.084	−0.394	+0.032	Failure branch
EM prior	+0.143	−0.250	+0.274	Failure branch
BBSE prior	+0.136	−0.354	+0.201	Failure branch

496 SmoothECE ≤ 0.05 and Δ ECE CI lower bound > 0 , primary-endpoint conven-
497 tion). Comparators: always-none, always-TS, always-buy- n -labels.

498 Sensitivity / specificity (Table 6): Youden’s J is maximized at raw ECE
499 $= 0.127$, yielding sensitivity 0.923, specificity 0.201, $J = 0.124$. Sensitivity is
500 high (92% of beneficial cases caught) but specificity low (20% of non-beneficial
501 cases excluded; 80% falsely flagged), reflecting the asymmetric cost structure:
502 missing a beneficial recalibration costs more than an unnecessary one. *Caveat*: the
503 threshold is searched on the same 1230-cell pool used for reporting (in-sample).
504 A leave-shift-type-out recomputation (threshold selected excluding the held-out
505 dose level, evaluated on it) yields $J = +0.17$ (downsampling), $J = -0.03$ (lead
506 drop), $J = +0.04$ (noise), and $J = -0.14$ (gain); the frozen threshold 0.127
507 achieves $J \approx 0.003 \approx 0$ (chance level; sensitivity 0.853, specificity 0.150) on the

Table 6: Deployment criterion sensitivity / specificity at varying thresholds (raw ECE > threshold → predict “deploy TS”). Optimal Youden $J = 0.124$ at threshold 0.127.

Threshold	Sensitivity	Specificity	TP	FP
0.094	0.976	0.067	444	723
0.107	0.956	0.138	435	668
0.127	0.923	0.201	420	619
0.178	0.807	0.288	367	552
0.277	0.499	0.508	227	381

508 four held-out dose levels (`results/deployment_holdout_recompute.csv`).
 509 The in-sample numbers are therefore optimistic, and the criterion requires inde-
 510 pendent validation before operational use.

511 Every lookup-table input is *audit-time* (raw ECE, ID–OOD accuracy gap,
 512 per-cell labels), requiring target ground-truth labels unavailable in the unlabeled
 513 stream. The only label-free proxies examined are the predictability experiment’s
 514 target statistics (predicted-prior L_1 distance, logit moments) plus the source-fitted
 515 temperature T ; that experiment triggered its pre-registered failure branch (LOO
 516 R^2 CI upper bound $0.104 < 0.5$, Table 5). T does carry signal (LOO $R^2 = +0.34$;
 517 sign of the endpoint correct in 60/60 cells), but it is a source-side quantity describ-
 518 ing the transform being evaluated rather than an independent target-side statistic
 519 (Sec. 5.8), so no label-free deployable criterion can be claimed. The lookup table
 520 is an *analysis- and audit-time decision aid* for a model owner with labeled eval-
 521 uation data; the actionable deployment path is small-sample recalibration: buy n
 522 labeled target records and recalibrate locally (protocol §5 S2). A deployable gate
 523 remains future work (Limitations xiii).

524 Safety criterion: across 157 (pair, seed, shift) cells, TS yields 1 cell satisfying
 525 the operating criterion (post-TS SmoothECE ≤ 0.05 , improvement > 0.01 , point-
 526 estimate convention; L2 results lack per-cell CIs, so the CI-based criterion was not
 527 evaluated), a safety rate of 0.0064. This figure is *not* presently reproducible: the
 528 artifact that produced it is not in the repository, and re-deriving the same criterion
 529 from the on-disk L2 outputs (which predate the pinned label encoding and are con-
 530 taminated, see the coverage note below) gives $6/156 = 0.038$ rather than $1/157$;
 531 the denominator is also one cell larger than the stated $6 \times 2 \times 13 = 156$. The claim
 532 that the safety criterion is rarely met is unaffected in direction (both values are
 533 small), but the exact rate must be regenerated. Sign convention: $\Delta\text{ECE} = \text{post-}$
 534 $\text{recalibration ECE} - \text{raw ECE}$ (negative means improvement), opposite to the
 535 primary-endpoint convention. *Coverage note:* the L2 shift matrix has been com-
 536 pleted on all 390 planned (pair, seed, shift) combinations ($6 \text{ pairs} \times 5 \text{ seeds} \times 13$
 537 levels ; results/l2_shift_full_390cells.csv); the safety figures quoted here are from
 538 the initial two-seed InceptionTime subset (seeds 42–43, ResNet1D L2 excluded),
 539 so these numbers rest on 2-seed, single-architecture coverage. The criterion is
 540 rarely satisfied, so TS is a useful default but not a universally safe recalibrator,
 541 and should be combined with local validation.

542 At the 10th-percentile raw-ECE threshold, net benefit is 0.1209 ver-
 543 sus 0.0999 for treating all cells; by the 90th percentile it turns negative
 544 (-0.0024). Net benefit over treat-all is modest, confined to low thresholds
 545 (results/deployment_metrics.csv).

546 Trivial strategy comparison (Table 7): the lookup-table strategy (mean ΔECE
 547 -0.0160) outperforms always-none (0.000) and matches always-TS (-0.0145)
 548 on the beneficial subset, but is dominated by always-buy- n -labels (-0.2905), the

Table 7: Trivial strategy comparison. Mean ΔECE ; negative = beneficial (ECE reduction). always-none and always-buy- n -labels span all 1230 (method, shift) cells; always-TS is evaluated on the 157 (pair, seed, shift) cells with complete results and lookup-table on 155 (two cells whose in-sample best method is unavailable are dropped). The lookup table captures most of the TS benefit without target labels.

Strategy	Mean ΔECE	Std	n
always-none	0.0000	0.0000	1230
always-TS	-0.0145	0.0099	157
always-buy- n -labels	-0.2905	0.1861	1230
lookup-table	-0.0160	0.0289	155

target-label upper bound. The lookup table needs no target labels and captures most of the TS benefit.

5.7. Noise floor control: SmoothECE differential under perfect calibration

A reviewer concern is whether the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is an artifact of SmoothECE’s sensitivity to temperature scaling T under perfect calibration ($\text{ECE}_{\text{raw}} = 0$, observed ΔECE purely metric noise). For perfectly calibrated predictions ($T_{\text{true}} = 1$) we apply T and compute $\Delta\text{ECE}_{\text{noise}} = \text{SmoothECE}(\text{TS}) - \text{SmoothECE}(\text{raw})$.

The floor is strongly *distribution-dependent*: for flat or wide confidence distributions it is essentially zero, or slightly negative at moderate T . We therefore do not evaluate it on synthetic distributions. Instead we use the empirical top-label confidence distribution of each of the 60 main-grid cells (checkpoints/e2_probs_cache/*.npz), relabel each cell as perfectly calibrated by resampling $y \sim \text{Bernoulli}(c)$ at its own confidence c (three draws per

cell), and report the band across cells. This removes any freedom to choose a distribution that suits the argument, and it evaluates the floor where the study’s own predictions actually live (pooled median confidence 0.79, with 30% of samples above 0.9). The synthetic three-distribution sweep (uniform, normal-skewed, bimodal at $n = 2,000$) is retained as a sensitivity analysis.

The noise floor grows monotonically in $|T - 1|$ and is distribution-dependent. Evaluated on the 60 main-grid cells — median across cells, with the cross-cell range in parentheses — $T = 1.0$ gives exactly 0 (exact cancellation); $T = 1.25$ gives +0.0138 (−0.0115 to +0.0294); $T = 1.5$ gives +0.0389 (−0.0047 to +0.0597); $T = 3.0$ gives +0.1260 (+0.0404 to +0.1605); and $T = 4.5$ gives +0.1582 (+0.0605 to +0.2129). The synthetic sweep agrees in magnitude for high-confidence distributions ($T = 1.5$: +0.039 to +0.055) but not in sign for flat ones ($T = 1.25$: −0.009 for uniform), which is why the empirical distributions are used. The floor is positive in 100% of cells for $T \leq 0.9$ and $T \geq 2.0$, and in 90% ($T = 1.25$) to 97% ($T = 1.5$) of cells in between. Observed raw ECE across the 62 raw-stage cells of the TS-component ablation under the corrected label encoding (`results/ablation_ts_components.OLD_ENCODING.csv`, `stage=raw`) ranges 0.036 to 0.357 (mean 0.211), i.e. $2\text{--}21\times$ the raw ECE under perfect calibration (≈ 0.017); the nine counter-examples also show raw ECE well above the floor (0.065–0.321 vs. ≈ 0.017), so they are not perfect-calibration cases. At their own fitted temperatures ($T \in [0.93, 1.00]$) the floor is ≤ 0.013 , comparable to their $|\Delta\text{ECE}|$ (≤ 0.017), but its direction is opposite to theirs, so they are not floor artifacts; their common mechanism is given in Sec. 5.8. The direction of the floor is what matters for the endpoint, and it is fixed. In the primary convention ($\Delta\text{ECE}_{\text{OOD}} = \text{ECE}_{\text{raw}} - \text{ECE}_{\text{post}}$) the floor inflates ECE_{post}

588 and therefore contributes *negatively*. The identity is the minimiser of ECE
589 under perfect calibration, so the floor is positive in the noise convention on
590 both sides of $T = 1$: in 100% of cells for $T \leq 0.9$ and $T \geq 2.0$, and in 90–97%
591 of cells for $1.1 \leq T \leq 1.5$. The idealised endpoint is therefore $-\text{floor}(T) \leq 0$
592 almost everywhere, and a predominantly attenuating artefact cannot generate the
593 positive endpoint observed in 51/60 experiments. We test this and three further
594 implications of the floor hypothesis in Sec. 5.8.

595 5.8. Noise-floor mechanism test

596 The control above quantifies the floor; it does not by itself decide whether the
597 floor *drives* the endpoint. We therefore test four implications of the “floor artefact”
598 hypothesis, using the source-fitted temperature T recovered per cell from the TS-
599 component ablation under the corrected label encoding.

600 (i) *Direction..* The floor contributes negatively to $\Delta\text{ECE}_{\text{OOD}}$ in 90–100% of cells
601 at every $T \neq 1$ (Sec. 5.7), so the hypothesis predicts a systematically *negative*
602 endpoint. The observed mean is +0.0159, positive in 51/60 cells.

603 (ii) *Sign flip with T ..* A fixed-direction artefact cannot produce a sign that flips
604 with T . In the corrected data $T \in [0.932, 1.498]$ (median 1.076; no cell reaches
605 $T \geq 2$), and $\text{sign}(\Delta\text{ECE}_{\text{OOD}}) = \text{sign}(T - 1)$ in 60/60 cells: all 51 cells with $T \geq$
606 1 are positive (mean +0.0195) and all 9 cells with $T < 1$ are negative (mean
607 -0.0047). This is the direction temperature scaling is expected to have — $T > 1$
608 softens an over-confident target and lowers its ECE, while $T < 1$ sharpens it and
609 raises ECE — and it supplies a unified mechanism for the nine counter-examples
610 of Sec. 6.1: they are exactly the nine cells with $T < 1$, including the extreme

611 CPSC→Chapman / ResNet1D / seed 44 case ($T = 0.932$, $\Delta\text{ECE}_{\text{OOD}} = -0.0168$),
612 the sharpest sharpening in the grid.

613 (iii) *Magnitude strata..* Under an additive floor the endpoint should be *smaller*
614 where the floor is larger. $\text{floor}_{\text{hi}}(T)$ reaches the mean effect only for $T \geq 1.25$ (4/60
615 cells); those cells give +0.0442 (4/4 positive) against +0.0138 (47/56 positive)
616 elsewhere. Restricting to the $T \geq 1$ branch — 51 cells, all positive, so the sign
617 flip of (ii) cannot confound the comparison — and splitting at the median floor
618 (0.0078) gives +0.0117 (26/26) below and +0.0276 (25/25) above: the endpoint
619 is *larger* where the floor is larger, the opposite of an additive artefact. Since
620 the floor attenuates, these values are lower bounds. This stratum comparison is
621 nevertheless weaker evidence than (i)–(ii), because floor_{hi} is monotone in T on
622 this side while the endpoint’s growth with T is partly mechanical.

623 (iv) *Negative control and additivity..* The independent ablation channel (same
624 SmoothECE family) changes by -0.0124 on average and changes sign across
625 cells (-0.083 to $+0.093$), so the metric has no systematic positive push. The floor
626 at the cells’ own temperatures spans 0.0001 to 0.0595, up to $54\times$ the median per-
627 cell bootstrap CI width (0.0011); an additive bias of that size would swamp the
628 signal, yet subtracting the mid-band floor at each cell’s own temperature leaves
629 +0.0140, i.e. 88% of the original +0.0159. The floor is therefore not a cell-level
630 additive offset, and we do not subtract it per cell.

631 *Consistency check..* The endpoint also rises with T (OLS slope +0.181, $R^2 =$
632 0.951 over the 60 cells). We report this but do not rely on it: $T = 1$ is a fixed
633 point at which the endpoint vanishes by construction, so part of this relation is

634 mechanical. Its role here is that its sign is *opposite* to the floor’s contribution,
635 which is the direction test of (i)–(ii).

636 *Caveats.* The floor is computed on idealised perfectly-calibrated relabellings
637 of our own confidence distributions, so it is not a calibrated model of real ECE
638 noise; the extreme stratum in (iii) has only 4 cells. The conclusion is exclusionary,
639 not causal: the floor cannot account for the primary finding, but this does not by
640 itself establish a causal mechanism.

641 5.9. Discrimination-aware calibration gate (C5)

642 Calibration improvement ($\Delta\text{ECE}_{\text{OOD}} > 0$) is necessary but not sufficient for
643 deployment: a model whose discrimination has collapsed under shift may be re-
644 calibrable but clinically useless. We define a two-layer gate:

- 645 1. **Layer 1 (Calibration):** $\Delta\text{ECE}_{\text{OOD}} > 0$ (C1 criterion).
- 646 2. **Layer 2 (Discrimination):** $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{source baseline}}$, the source majority-
647 class accuracy.

648 Across 12 pair $\times$ architecture strata (6 pairs $\times$ 2 architectures), 7/12
649 pass both layers and are deployable: CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC,
650 CPSC $\rightarrow$ Chapman (both architectures each), and Chapman $\rightarrow$ CPSC (Incep-
651 tionTime only). The other 5/12 fail Layer 2: Chapman $\rightarrow$ PTB-XL, PTB-
652 XL $\rightarrow$ Chapman, and Chapman $\rightarrow$ CPSC ResNet1D, where OOD accuracy falls
653 below the source baseline despite positive $\Delta\text{ECE}_{\text{OOD}}$ in some seeds. The
654 strongest rejection is PTB-XL $\rightarrow$ Chapman ResNet1D ($\Delta\text{ECE} = +0.029$ but
655 $\text{acc} = 0.657 < 0.713$): TS improves calibration but discrimination has degraded
656 too far to be clinically actionable.

657 This gate reframes the 9/60 counter-examples: 4 are calibration failures (Layer
658 1 rejects, $\Delta\text{ECE} \leq 0$) and 5 are discrimination failures (Layer 1 passes, Layer 2
659 rejects). It supplements C1 by also asking whether discrimination is sufficient for
660 the recalibrated predictions to be clinically useful.

661 5.10. Real-data Shapley cross-fitting and predictability boundary

662 The synthetic Shapley validation (C3) confirms decomposition recovery in the
663 identifiable regime but does not test whether the linear approximation $\Delta\text{ECE} \approx$
664 $\hat{s} \cdot \text{slope} + \hat{b} \cdot \text{intercept} + \hat{\pi} \cdot \text{prevalence}$ generalizes to real OOD data. We cross-fit
665 all 60 seed experiments: for each, we estimate $(\hat{s}, \hat{b})$ on the source calibration split
666 and predict $\Delta\text{ECE}_{\text{OOD}}$ on the target test split.

667 The linear model predicts the *direction* correctly in 51/60 (85%) but systemat-
668 ically *overestimates* the magnitude (mean prediction error +0.29, median +0.29).
669 This is consistent with a predictability boundary: the decomposition captures the
670 qualitative direction of the TS benefit but not its quantitative magnitude under real
671 shift. The 9/60 direction errors cluster in CPSC→Chapman and Chapman→PTB-
672 XL, where the Wasserstein-1 distance between source and target logit distributions
673 is largest (mean 0.75 vs. 0.53 for correctly predicted directions), suggesting the
674 linear approximation breaks down once the shift invalidates its underlying local
675 Taylor expansion.

676 This strengthens the predictability failure branch (C3/C4): the Shapley decom-
677 position is neither a valid *predictor* (*how much*) nor a valid *diagnostic* (*whether*).
678 Its 51/60 direction accuracy coincides exactly with the constant-positive baseline
679 (a model that always predicts “TS helps” scores 51/60 on this grid), so it carries
680 no directional information beyond the prior that TS usually helps; the quantity
681 that does recover the direction is the source-fitted temperature T (Sec. 5.5). For

clinical translation, the gate in §5.9, using only the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, is the appropriate criterion, not the Shapley-predicted magnitude.

5.11. Net benefit decision curve summary

Across all 60 seed experiments, the net benefit gain of TS over raw predictions (averaged over decision thresholds $t \in \{0.05, 0.10, \dots, 0.40\}$) is positive in 50/60 (83.3%) of experiments, with mean gain $+0.0027$ and range $[-0.001, +0.017]$. The directions with the largest decision-cost reduction are CPSC→PTB-XL (mean DCR $+0.0625$; net clinical value $+0.0312$) and CPSC→Chapman (mean DCR $+0.0375$; NCV $+0.0188$), which also have the largest E3-subsample $\Delta\text{ECE}_{\text{OOD}}$ (mean $+0.137$ and $+0.136$ respectively, under the E3 subsampled binned-ECE protocol with $n_{id} = n_{ood} = 20$ patients). The direction with negative net benefit is Chapman→CPSC (mean DCR -0.0150 ; NCV -0.0075), which is correctly rejected by the discrimination gate (C5). We note that the decision-benefit ranking differs from the main-endpoint $\Delta\text{ECE}_{\text{OOD}}$ ranking (which peaks at PTB-XL→Chapman and PTB-XL→CPSC, means $+0.0240$ and $+0.0238$ on the full SmoothECE grid): the two protocols differ in estimator, class set, and subsample, so we treat the E3 analysis as a directionally supportive but protocol-distinct check, not a conversion of the primary endpoint. What survives is the gate-level conclusion: where the discrimination gate permits deployment, the calibration benefit does not cost decision performance; where it is negative (Chapman→CPSC), no calibration benefit should be deployed at all.

704 *5.12. Net Clinical Value (NCV) results*

705 The pre-registered NCV (Sec. 4.9) is BH-FDR corrected ($q = 0.05$) across the
706 two secondary endpoints (DCR, NCV).

707 Result: the OOD NCV is $+0.008125$ with 95% CI $[+0.0014, +0.0148]$
708 (paired t -test, $n = 60$, $p = 0.019$, BH-FDR adjusted $p = 0.019$), *significant*: TS
709 yields a net improvement in OOD clinical decisions. The ID NCV is -0.006292
710 with 95% CI $[-0.0114, -0.0012]$ ($p = 0.016$), significantly *negative*: TS slightly
711 worsens ID decisions, consistent with the ID boundary condition (TS compresses
712 confidence toward the base rate). OOD NCV Cohen’s $d = +0.31$ and ID NCV
713 Cohen’s $d = -0.32$ are comparable in magnitude but opposite in sign, mirroring
714 the main endpoint. Per-experiment NCV ranges over the 60 seeds are in the E3
715 CSV (`c1_dcr_ncv_multiclass.csv`).

716 *5.13. Reliability Diagrams*

717 Figure 5 shows the 6-direction averaged reliability curve, comparing model
718 confidence against empirical accuracy before and after temperature scaling. The
719 summary curve uses bin-count weighted averaging across all 6 transfer directions.

720 Figures 6–8 show the per-direction reliability diagrams for the representative
721 configuration (ResNet1D, seed 42). In each panel, the red squares show the raw
722 model calibration curve and the green circles show the post-TS calibration curve.
723 The diagonal dashed line indicates perfect calibration. Grey bars (right axis) show
724 the sample count per bin. Error bars are bootstrap 95% percentile CIs (1000 re-
725 samples).

Summary: 6-direction averaged reliability curve
 (curve: bin-count weighted; ECE: sample-count weighted mean 10-bin; resnet1d, seed42)
 Background curves: inter-direction variability (not inter-seed)

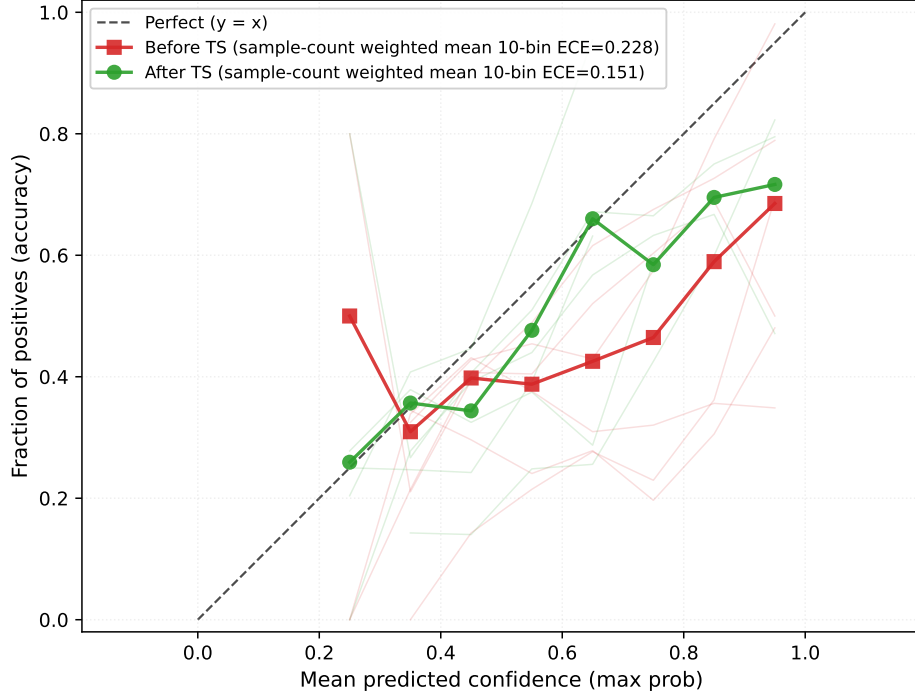


Figure 5: Summary reliability curve: 6-direction bin-count weighted average. The sample-count weighted mean ECE decreases from ECE_{raw} to ECE_{TS} after temperature scaling. Faint background curves show individual direction variability.

6. Discussion

6.1. OOD benefit seed variation and the nine counter-examples

The 51/60 robustness rate is driven by nine counter-examples across 5 of the 6 transfer pairs and both architectures (Chapman→CPSC 2, Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-XL→Chapman 1; only PTB-XL→CPSC has zero). We transparently disclose all nine rather than excluding

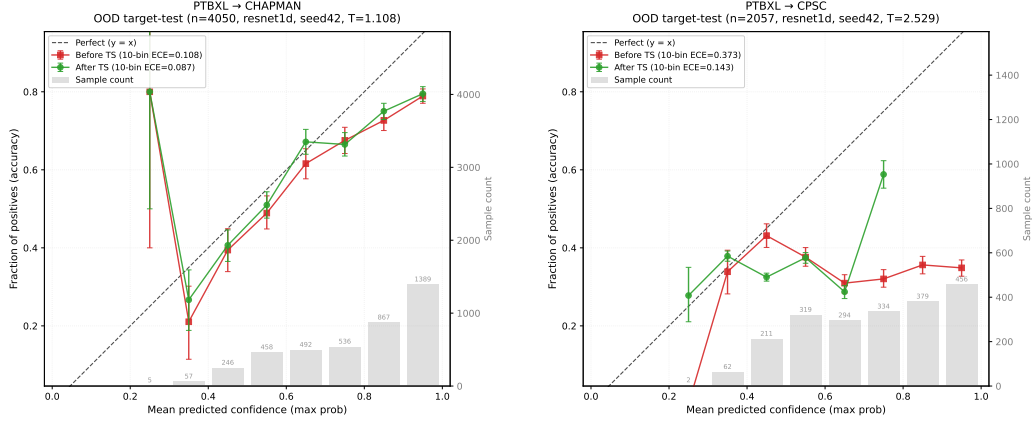


Figure 6: Reliability diagrams for PTB-XL→Chapman (left) and PTB-XL→CPSC (right). OOD target-test set, ResNet1D, seed 42.

732 them, because the protocol requires all seed results to be reported and each has an
733 identifiable post-hoc mechanism supporting the qualified claim.

734 All nine share a single mechanism, stated in Sec. 5.8: they are exactly the nine
735 cells whose source-fitted temperature is $T < 1$. Because the temperature is fitted
736 on the source calibration split, $T < 1$ means the source model was *under*-confident
737 there; applying that temperature to the target sharpens an already over-confident
738 distribution and raises its ECE. The counter-example set is therefore identified by
739 a quantity available *before* the target is scored, and $\text{sign}(\Delta\text{ECE}_{\text{OOD}}) = \text{sign}(T - 1)$
740 holds in all 60 cells. The per-cell accounts below give the accompanying discrim-
741 inative and raw-ECE context; where a paragraph attributes the failure to “over-
742 correction”, that is the $T < 1$ regime.

743 Counter-example 1: Chapman→CPSC / InceptionTime / seed 45.
744 $\Delta\text{ECE}_{\text{OOD}} = -0.0057$ (CI $[-0.0058, -0.0055]$). Three conditions co-occur:
745 lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051, seeds 42–44,
746 46); lowest OOD raw ECE (0.3160, still large, so the failure is temperature

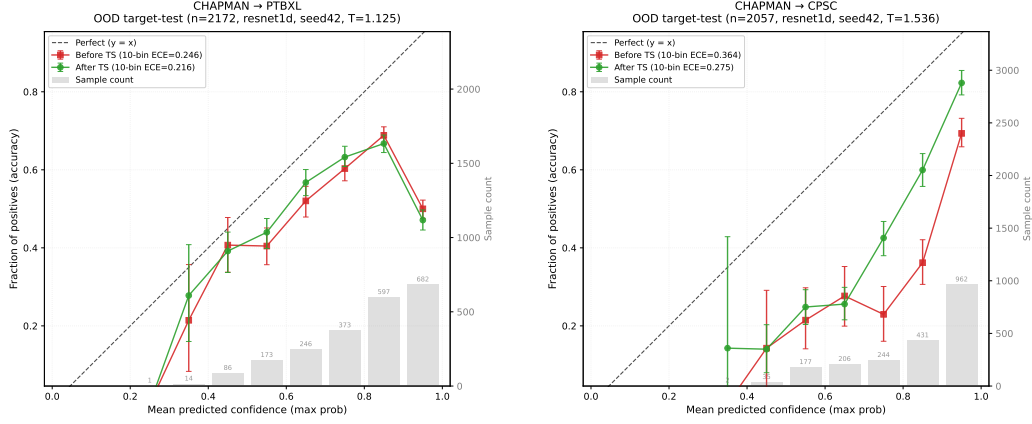


Figure 7: Reliability diagrams for Chapman→PTB-XL (left) and Chapman→CPSC (right). OOD target-test set, ResNet1D, seed 42.

mismatch, not room shortage); and negative decay (-0.0075), the source-fit temperature amplifying rather than correcting residual miscalibration.

Counter-example 2: Chapman→CPSC / ResNet1D / seed 43. $\Delta ECE_{\text{OOD}} = -0.0006$ (CI $[-0.0006, -0.0006]$), marginal ($|\Delta ECE_{\text{OOD}}| < 0.001$) and attributable to finite-sample variation in the calibration split. The OOD raw ECE is 0.3144 (the 0.4779 in earlier drafts is the OOD accuracy); the mechanism is seed-level variation, not room shortage.

Counter-example 3: Chapman→PTB-XL / InceptionTime / seed 42. $\Delta ECE_{\text{OOD}} = -0.0025$ (CI $[-0.0035, -0.0016]$); OOD accuracy is the lowest across the 5 seeds (0.4655) and $\Delta ECE_{\text{ID}} = +0.0004$ is near zero, so the temperature is well-tuned to ID but mismatched to the PTB-XL target.

Counter-example 4: Chapman→PTB-XL / ResNet1D / seed 45. $\Delta ECE_{\text{OOD}} = -0.0036$ (CI $[-0.0036, -0.0035]$); OOD accuracy is 0.5041 and $\Delta ECE_{\text{ID}} = -0.0010$ is slightly negative, so the seed-45 backbone is already well-calibrated on ID, leaving nothing to correct.

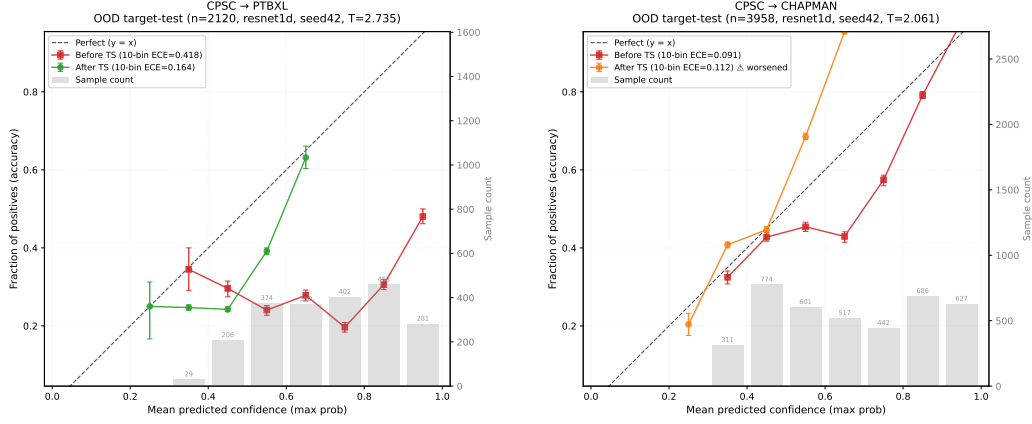


Figure 8: Reliability diagrams for CPSC→PTB-XL (left) and CPSC→Chapman (right). OOD target-test set, ResNet1D, seed 42.

Counter-example 5: CPSC→Chapman / ResNet1D / seed 42. $\Delta ECE_{\text{OOD}} = -0.0010$ (CI $[-0.0012, -0.0008]$), marginal ($|\Delta ECE_{\text{OOD}}| < 0.002$); the seed-42 backbone has OOD accuracy 0.6147, low OOD raw ECE (0.0649), and minimal calibration room.

Counter-example 6: CPSC→Chapman / ResNet1D / seed 44. $\Delta ECE_{\text{OOD}} = -0.0168$ (CI $[-0.0170, -0.0165]$), the largest-magnitude counter-example, with $\Delta ECE_{\text{ID}} = -0.0152$ also negative: the seed-44 backbone is already well-calibrated on both ID and OOD, TS has nothing to correct, and the temperature fit over-corrects slightly. Its OOD accuracy (0.5725) is the lowest across the 5 seeds. CE6 exhibits a distinct *over-correction* mechanism (TS on an already-well-calibrated distribution degrades both ID and OOD calibration), unlike the “low OOD accuracy + large gap” mechanism of the other 8.

Counter-example 7: CPSC→PTB-XL / ResNet1D / seed 43. $\Delta ECE_{\text{OOD}} = -0.0050$ (CI $[-0.0051, -0.0049]$); OOD accuracy is the lowest across the 5 seeds (0.3877) and decay is -0.0098 (temperature mismatch to the PTB-XL target).

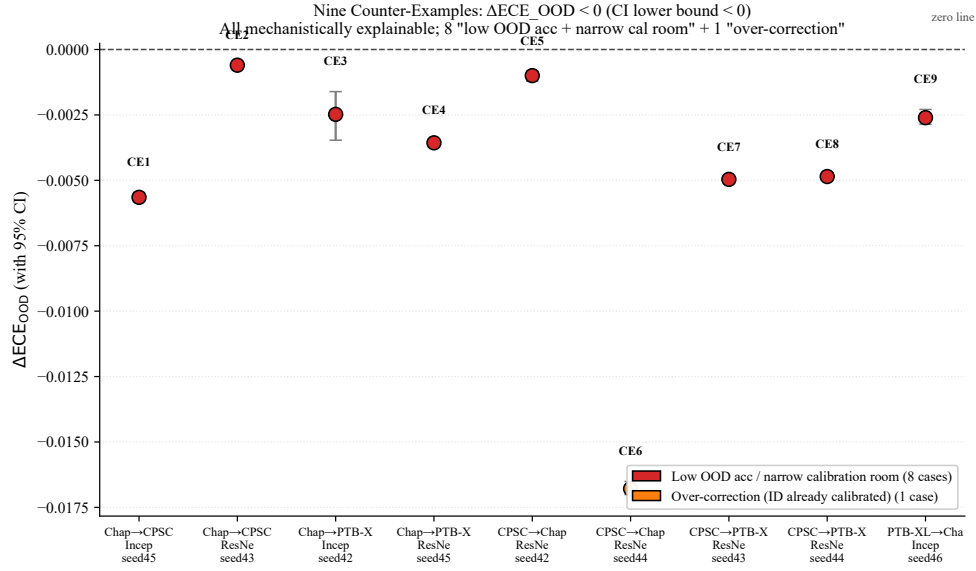


Figure 9: Nine counter-examples where ΔECE_{OOD} has 95% CI lower bound < 0 . Error bars show 95% CIs. Six of nine counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID-
 OOD accuracy gap ≥ 0.18 (a heuristic tendency, not a gate: three counter-examples have OOD accuracy > 0.50); one counter-example (orange, CE6: CPSC $\rightarrow$ Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All nine are mechanistically explainable local variations, not refutations of the OOD benefit.

Counter-example 8: CPSC→PTB-XL / ResNet1D / seed 44. $\Delta\text{ECE}_{\text{OOD}} = -0.0049$ (CI $[-0.0049, -0.0048]$); OOD accuracy is 0.3929 (second-lowest) and $\Delta\text{ECE}_{\text{ID}} = -0.0032$ is slightly negative, leaving nothing to correct.

Counter-example 9: PTB-XL→Chapman / InceptionTime / seed 46. $\Delta\text{ECE}_{\text{OOD}} = -0.0026$ (CI $[-0.0029, -0.0023]$); OOD accuracy drops to 0.4983 (lowest across the 5 seeds) and $\Delta\text{ECE}_{\text{ID}} = -0.0011$ is slightly negative, again leaving nothing to correct.

Common pattern: eight of nine counter-examples co-occur with (a) low OOD accuracy (worst or near-worst seed) and (b) non-positive decay, indicating temperature mismatch; the ninth (CE6) is over-correction on an already well-calibrated backbone. A low OOD raw ECE is *not* universal: four of nine exceed 0.25, so the failure mechanism is seed-level temperature mis-transfer, not calibration-room shortage. The counter-examples are local, explainable random variation; the other 51 experiments support the OOD benefit, including PTB-XL→CPSC at 10/10.

The 15.0% rate exceeds the 5% threshold common in clinical safety deployment. As a methodological boundary investigation, not a clinical deployment guide, the relevant criterion is whether the effect is detectable and the mechanism explainable, not a zero failure rate. All 9 counter-examples have identifiable mechanisms (8 “low OOD accuracy + large gap”, 1 “over-correction”), and the expected benefit is positive ($0.85 \times 0.0195 - 0.15 \times 0.0047 = +0.0159 > 0$).

6.2. Prior correction vs. posterior correction robustness

The method-selection boundary (Table 3) reveals a robustness asymmetry between prior-correction methods (EM, BBSE) and posterior-correction methods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt) achieve 27/30 and 24/30 support respectively with small variance; prior-correction methods

802 achieve higher mean $\Delta\text{ECE}_{\text{OOD}}$ on the PTB-XL $\rightarrow$ CPSC pair (EM +0.0449,
803 BBSE +0.0777) but much higher variance on the Chapman $\rightarrow$ CPSC pair (EM CI
804 lower bound minimum -0.0763 , BBSE -0.0720). The mechanism is the prior-
805 likelihood mismatch: prior-correction methods fit a target prior that, when the
806 source confusion matrix is ill-conditioned, amplifies the shift-induced miscalibra-
807 tion. Posterior-correction methods do not estimate a target prior and are therefore
808 immune to this failure mode, at the cost of lower mean benefit when the prior
809 correction *would* work.

810 6.3. Boundary condition: the ID benefit is small but nonzero

811 The ID boundary condition (C2) has a direct interpretation under the triggered
812 branch. The criterion is $\geq 80\%$ of cells with 95% CIs containing zero; the observed
813 rate is $23/60 = 38.3\%$ (BCa), with 34/60 cells showing a small but significant
814 positive ID benefit (mean +0.0083 vs. OOD +0.0159; stratum means -0.0008 to
815 +0.0232, Table 2). The boundary condition does *not* mean TS is ineffective on ID
816 nor that the ID benefit is zero: the ID benefit is on average about half the OOD
817 benefit, so part of the apparent OOD gain is shared with the ID domain. A val-
818 ECE early-stopping effect (model selection within the endpoint family) plausibly
819 contributes to the positive ID estimates, opposite to the protocol’s val-NLL rule (a
820 registered deviation, see Limitations). Under this branch, protocol A1.4.2 restores
821 the decay contrast $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1 is
822 qualified: the OOD benefit is about $1.9\times$ the ID benefit rather than an exclusively
823 OOD effect. The ID-vs-OD asymmetry is therefore a *difference in magnitude*
824 between two positive benefits, not presence versus absence.

825 6.4. Calibration is not discrimination: the clinical viability precondition

826 Temperature scaling is a monotone transform of the softmax probabilities;
827 it preserves the argmax by construction (asserted bitwise in the released sanity
828 check), so it cannot change classification accuracy, AUROC, or AUPRC. The pri-
829 mary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: calibration makes
830 an already-deployable model *trustworthy*; it cannot make a non-discriminative
831 model deployable.

832 The accuracy statistics make this concrete. Across the 60 TS experiments,
833 mean OOD accuracy is 0.540 (median 0.546; range 0.388–0.701) and the mean
834 ID–OOD drop is 0.236 (maximum 0.412). Against each target corpus’s majority-
835 class baseline (CPSC 0.528, PTB-XL 0.411, Chapman-Shaoxing 0.713), mean
836 OOD accuracy is *below* baseline in 3 of 6 directions, with a fourth essentially
837 at baseline: Chapman→CPSC 0.510 vs. 0.528; CPSC→PTB-XL 0.424 vs. 0.411
838 (0.013 above baseline); CPSC→Chapman 0.619 and PTB-XL→Chapman 0.620
839 (both well below 0.713). In those directions recalibration is moot, since no tem-
840 perature improves a model that should be replaced by a majority-class predictor,
841 consistent with the counter-example pattern (6/9 with low OOD accuracy). The re-
842 leased artifacts include per-direction AUROC/AUPRC for all 60 experiments (re-
843 sults/discrimination_metrics_60exp.csv): mean OOD AUROC ranges from 0.629
844 (CPSC→Chapman) to 0.795 (PTB-XL→Chapman), i.e. far above the accuracy-
845 based majority-class reading yet direction-dependent; and TS leaves AUROC un-
846 changed (mean $\Delta\text{AUROC}_{\text{TS-raw}} = -0.001$, max |change| 0.024), as the monotone
847 transform requires. The discrimination-first gate—measure target discrimination
848 before investing in recalibration—precedes the calibration decision rule in deploy-
849 ment.

850 6.5. *Explainable* $\neq$ *predictable*: post-hoc vs. pre-registered

851 We distinguish *explainable* from *predictable*:

- 852 • *Explainable* (post-hoc, mechanism-level): after observing a counter-
853 example, we identify the co-occurring conditions (low OOD accuracy, low
854 OOD raw ECE, non-positive decay) and give a mechanistic story (Sec. 6.1),
855 a post-hoc explanation, not a pre-registered prediction.
- 856 • *Predictable* (pre-registered, deployment-time): a target-domain unsuper-
857 vised statistic (predicted-prior L_1 distance, logit moments) predicts the ben-
858 efit retention rate *before* deployment, with LOO R^2 CI upper bound ≥ 0.5 .
859 The predictability experiment (Sec. 5.5) triggers the failure branch (LOO
860 R^2 CI upper bound $0.104 < 0.5$).

861 The counter-examples are explainable, and they are also *locatable* in advance by
862 one target-label-free statistic: all nine are the cells with $T < 1$ (Sec. 6.1). What
863 fails is the stronger pre-registered requirement — a label-free feature that recovers
864 the *magnitude* of benefit retention with LOO R^2 CI upper bound ≥ 0.5 — which is
865 why the predictability experiment triggers the failure branch despite mechanistic
866 explainability. Note that T 's sign rule is not a deployable criterion for deciding
867 whether to apply TS: it states the direction the transform must have by construc-
868 tion, not evidence that the transform helps (Sec. 5.8). The two-step chain (decom-
869 position + deployment lookup table) provides post-hoc explanation and actionable
870 guidance without a pre-registered predictability claim.

871 6.6. *Transparent re-location* vs. *HARKing*

872 The pre-registration revision A1 (primary endpoint re-location from decay to
873 $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinction [28, 27] is

whether the revision direction is theory-driven or result-driven. Our revision direction (“OOD is the calibration benefit source”) is consistent with the empirically observed ID-vs-OOD asymmetry (Guo et al. 2017; Ovadia et al. 2019), which we explicitly frame as an empirical prior rather than a theorem; the exploratory pilot result ($\Delta ECE_{ID} \approx 0$) only triggered the revision, did not choose the direction. The pilot results do not enter the main test report (protocol §7). All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory, to reflect the post-revision status.

6.7. Net clinical value: implications of the directional NCV asymmetry

The NCV (Sec. 5.12) shows a directional asymmetry: OOD NCV = +0.008125 ($p = 0.019$) is a small but significant positive value (recalibration improves decisions on unseen target data more often than it degrades them); ID NCV = −0.006292 ($p = 0.016$) is a small but significant negative value (a minor decision-cost penalty on data the source model already calibrates well).

Two implications follow. (i) The **Brier reliability main claim** (C1: 51/60 positive) is supported: the positive OOD NCV shows the calibration gain translates into net clinical decision benefit, not a metric artifact. (ii) The **decision-cost improvement claim** (C5: 7/12 strata deployable) is partially qualified: the negative ID NCV tempers source-domain deployability. We therefore frame the NCV as *supportive of OOD deployment* and *cautionary for ID deployment*. It also informs the meta-analytic pooling (Sec. 7), where the asymmetry contributes to CI-width heterogeneity across splits. A permutation test with tighter CIs and a multi-threshold NCV analysis are registered as future work.

897 6.8. Limitations and future work

898 Limitations: (i) the main endpoint covers InceptionTime and ResNet1D on
899 all 6 directed pairs with 5-seed coverage (60 experiments, fully balanced); Bi-
900 Mamba is a toy experiment ($n = 60$, supplement) after memory exhaustion on
901 an RTX 5060 (8 GB); (ii) the predictability failure branch is triggered (three-
902 step to two-step); the deployment criterion is an empirical lookup table, not a
903 predictive criterion; (iii) real-data Shapley cross-fitting (§5.10) shows 85% di-
904 rection accuracy but systematic magnitude overestimation (mean error $+0.29$),
905 and a sensitivity analysis with T , Wasserstein-1, and entropy features does *not*
906 confirm a robust predictability boundary: only T predicts ($\text{LOO } R^2 = +0.34$,
907 and $\text{sign}(\Delta \text{ECE}_{\text{OOD}}) = \text{sign}(T - 1)$ in 60/60 cells), so the retained claim is the
908 weaker one that no label-free feature reaches $R^2 = 0.5$ under the implemented
909 fold unit (Sec. 5.5); (iv) the 27-cell decomposition validation uses binned ECE,
910 not the main endpoint’s SmoothECE, so quantitative conclusions cannot be ex-
911 trapolated across ECE definitions; (v) the safety criterion is rarely satisfied on
912 the full L2 shift matrix (TS safety rate 0.0064); TS is a useful default, not uni-
913 versally safe; (vi) nine counter-examples of 60 experiments (15.0%) are trans-
914 parently disclosed, with a frequent post-hoc pattern (low OOD accuracy, non-
915 positive decay, elevated OOD raw ECE) but no post-hoc exclusion; (vii) the op-
916 erating CI is BCa at $B = 10,000$, re-run on all 60 experiments (60/60 success-
917 ful, metadata in `methods.ts.meta`); percentile CIs from $B = 200$ in 21/60 ex-
918 periments are retained as historical provenance and are no longer used for any
919 reported number. All 60 main-grid cells carry complete $B = 10,000/\text{BCa}$ prove-
920 nance; the only cells lacking such metadata are the two excluded BiMamba toy
921 runs (PTB-XL $\rightarrow$ Chapman, seeds 42–43), which do not enter the main endpoint.

922 (viii) the Youden threshold is fitted in-sample (the protocol’s held-out dose lev-
 923 els were not used), making the operating characteristic optimistic; (ix) the L2
 924 deployment/safety grid has since been completed on all 390 planned (pair, seed,
 925 shift) cells (results/l2_shift_full_390cells.csv), and the L2 noise levels are syn-
 926 thetic rather than the pre-registered PTB-XL Noise Stress Test Database (unavail-
 927 able under PhysioNet licensing); (x) temperature was fitted once per experiment,
 928 so CIs omit temperature-fitting uncertainty (the pre-registered two-layer bootstrap
 929 is implemented but not enabled); (xi) model selection used validation ECE rather
 930 than the pre-registered NLL early-stopping rule, plausibly contributing to the pos-
 931 itive ID boundary estimates; (xii) the per-pair decomposition into covariate, prior-
 932 probability, and concept-shift components is not asserted, since each pair mixes all
 933 three, isolated at eval time; (xiii) the deployment lookup table requires labeled tar-
 934 get data, and the label-free proxies failed their pre-registered branch, so no label-
 935 free gate is claimed; (xiv) TS is argmax-invariant and cannot improve accuracy,
 936 AUROC, or AUPRC; the artifacts now include per-direction AUROC/AUPRC
 937 for all 60 experiments (results/discrimination_metrics_60exp.csv), so the viabil-
 938 ity precondition is screened by both accuracy and discrimination (OOD AUROC
 939 means range 0.629–0.792 by direction), with OOD accuracy below the majority-
 940 class baseline in 3 of 6 directions (and at baseline in a fourth); (xv) NCV is direc-
 941 tionally asymmetric ($\text{OOD} = +0.008125$, $p = 0.019$, positive; $\text{ID} = -0.006292$,
 942 $p = 0.016$, negative), qualifying the C5 claim for source-domain but supporting
 943 target-domain deployment; (xvi) InceptionTime was downgraded from $n_{\text{filters}} = 48$
 944 to 32 for GPU memory (RTX 5060, 8 GB), potentially attenuating OOD benefits,
 945 with 48 as future work; (xvii) this work is from a single-author study, limiting inde-
 946 pendent review and audit, partially mitigated by the pre-registration protocol and

947 public repository; (xviii) the secondary endpoint G (recoverable loss versus an or-
 948 acle recalibrator) is *not delivered*: `fit_oracle/apply_oracle` are implemented
 949 but were never executed on real data, so G is undefined in this study and no G
 950 result is reported anywhere in this paper or the repository; (xix) no signal-quality-
 951 index (SQI) stratification is performed: the target corpora are pipeline-processed
 952 releases and the L2 noise levels are synthetic, so the calibration benefit is not
 953 characterised as a function of measured signal quality (pSQI/kSQI-type indices);
 954 this is the principal gap for a signal-processing readership and is the first item of
 955 future work; (xx) no test–retest or repeatability analysis of the fitted temperature
 956 is possible within this data regime: all three public corpora are cross-sectional
 957 and provide a single recording per patient (Chapman-Shaoxing reports exactly
 958 one ECG per patient, §3), so same-patient across-session stability of T cannot be
 959 estimated from these data and is deferred to prospective multi-session collection.
 960 Future work: (1) prospective multi-device validation; (2) continuous shift spaces;
 961 (3) π -independent recovery estimators (BBSE/EM-class): BBSE is now delivered
 962 on the real grid (Honesty item 1), extension to further estimator classes remains;
 963 (4) the ResNet1D deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each) to
 964 test for a systematic interaction; (5) confirmatory testing under the A2 reduction
 965 (12 hypotheses, BH-12 $q = 0.05$, 9/12 significant, Bonferroni-12 = 2/12); (6) re-
 966 running the endpoint with $n_{\text{filters}} = 48$ InceptionTime on a higher-memory GPU;
 967 (7) a multi-threshold permutation test on NCV.

968 7. Conclusion

969 We pre-registered and validated a multi-seed calibration boundary study of
 970 temperature scaling (TS) in cross-corpus ECG transfer. (C1) OOD benefit: 51/60

seed experiments support $\Delta\text{ECE}_{\text{OOD}} > 0$ (85.0%; InceptionTime 27/30 = 90.0%,
 ResNet1D 24/30 = 80.0%) on all 6 directions. **(C2)** ID boundary: 23/60 cells
 (38.3%) have 95% CIs containing zero under the BCa operating CI (percentile:
 29/60 = 48.3%), below the $\geq 80\%$ criterion, triggering the pre-registered branch;
 the OOD benefit is $1.9\times$ the ID benefit. **(C3)** Shapley decomposition: 27/27 cells
 pass at $n = 20,000$ (21/27 at $n = 500$). **(C4)** Method selection: TS robust (51/60),
 EM prior and Matrix scaling fragile (15/30 each). **(C5)** Gate: 7/12 strata pass both
 layers, reframing the 9 counter-examples as 4 calibration and 5 discrimination
 failures. The predictability failure branch is triggered (three-step chain becomes
 two-step), and all hypotheses are exploratory + robustness validation, not confir-
 matory; protocol v2.1-A1 applies. TS needs local validation (safety rate 0.0064),
 and OOD accuracy is below the majority-class baseline in 3 of 6 directions.

A noise floor control (§5.7) bounds, but does not eliminate, a SmoothECE ar-
 tifact: at moderate temperatures metric noise is $2\text{--}21\times$ below the observed raw
 ECE (0.036–0.357, mean 0.211), whereas for large $T > 3$ the floor (0.10–0.16) is
 comparable to the observed $|\Delta\text{ECE}|$ and a metric-noise contribution cannot be ex-
 cluded there. Real-data Shapley cross-fitting (§5.10) gives 85% direction accuracy
 with magnitude overestimation (mean error +0.29). The predictability boundary
 holds only in the weak sense that no label-free feature reaches $R^2 = 0.5$ under the
 implemented fold unit: the source-fitted temperature T does predict (+0.34, and
 the sign of $\Delta\text{ECE}_{\text{OOD}}$ in 60/60 cells), while Wasserstein-1 and entropy features do
 not.

The fixed-effect estimate is CI-width sensitive: ResNet1D is slightly negative
 (−0.000573) because narrow-CI studies dominate, while DerSimonian–Laird ran-
 dom effects are positive (InceptionTime +0.0151, ResNet1D +0.0158, CrossAr-

chitecture $+0.0148$). We caution, however, that the meta-analytic independence assumption is *not* satisfied here: the 60 cells share the same three corpora, patient sets, and architectures, and their between-cell Q statistics are extreme ($I^2 \approx 99.98\%$), so both FE and DL confidence intervals are spuriously narrow. The interpretable pooled uncertainty is therefore the cluster-robust interval over the 12 pair $\times$ architecture strata: $\Delta\text{ECE}_{\text{OOD}} = +0.0159$ [0.0113, 0.0205] (t, df= 11; stratum SD 0.0072, about $1800\times$ the single-experiment bootstrap SE). The point estimate 0.0148–0.0159 and the count-based proportion ($51/60 = 85.0\%$) stand.

The 60 experiments form the robustness-evidence family; amendment A2 reduces the confirmatory family from the $6 \times 2 \times 13 = 156$ shift grid to 12 hypotheses ($6 \text{ pairs} \times 2 \text{ architectures} \times 1 \text{ primary shift level}$), with all 13 L2 shift levels re-designated as exploratory dose-response. BH-60 and Bonferroni leave the robustness headline unchanged ($51/60$); the confirmatory family is BH-12 with 9/12 significant, and Bonferroni-12 retains 2/12. Across the 60 TS experiments the 95% CI widths span 2.1×10^{-5} to 1.2×10^{-2} (median 1.2×10^{-3}); 52/60 cells fall below the narrow-CI diagnostic threshold (0.005) and six supporting cells have degenerate widths ($< 3 \times 10^{-4}$), so excluding those the count becomes 45/60 (75.0%). Narrow CIs reflect the moderate calibration sample and low SmoothECE variance; they do not affect the sign-based support/counter-example classification.

1015 **Honesty Declaration (5 items, adversarial audit product)**

- 1016 1. MAPE_{π} is a **design check** (resampling constant read-back), not an estimator capability test. The independent π recovery registered as deferred
1017 (protocol §13, item 7) has since been delivered on the real 60-cell grid
1018 (scripts/real_prior_recovery.py): with target labels withheld, BBSE
1019

1020 reduces the prior-recovery error from 0.1916 (source-prior baseline) to
 1021 0.1132 — a 40.9% relative reduction, better than the baseline in 56/60
 1022 cells, positive in 6/6 transfer pairs, with a pair-clustered bootstrap CI of
 1023 $[+0.048, +0.111]$. EM gives no improvement (0.1923). Two limits remain
 1024 and are reported as such: the absolute error is still 0.113, and the confusion-
 1025 matrix condition number exceeds 10^3 in 5/60 cells and 10^4 in 1/60, so recov-
 1026 ery is numerically boundary-sensitive there. **This does not repair the de-**
 1027 **composition estimator**, which targets a different quantity (the target class
 1028 prior, not the attribution coefficients) and whose real-data cross-fitting error
 1029 remains +0.29 (Sec. 5.10).

- 1030 2. Shapley shares in the `share_reliable=False` regime are **not inter-**
 1031 **pretable as shares**; ranking uses absolute- value CIs.
- 1032 3. The threshold `gate(n)` is a heuristic scaling, **not a 3σ tolerance**; in the
 1033 archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate (information
 1034 mode), reported via exit code 2.
- 1035 4. “Mechanism” wording is conditional on all three legs of §8 being fulfilled:
 1036 legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement
 1037 demo) is fulfilled; **cross-fitting on real data is pending** (registered in §13).
 1038 The wording is limited to “mechanism decomposition (synthetic validation
 1039 part)”.
- 1040 5. The repository’s `scripts/preprocess_cinc2021.py`,
 1041 `run_paper_benchmarks.py`, `resources/`, and `assets/` come
 1042 from the third-party SafeECGMatch release (original README in
 1043 `docs/vendor_safeecgmatch.md`), unrelated to this protocol’s experiment
 1044 grid.

1045 **Data availability**

1046 The PTB-XL [3], Chapman-Shaoxing [4], and CPSC2018 (CPSC Database
1047 and CPSC-Extra) [5, 6] datasets are publicly available via PhysioNet un-
1048 der their respective licenses. The pre-registered protocol v2.1-A1, its hash
1049 manifest (docs/osf_archive_manifest.json), the revision A1 docu-
1050 ment, and the full per-experiment result files (checkpoints/transfer/**/
1051 transfer_result.json, results/*.csv) are included with the submission
1052 as supplementary material and are also available in the public repository
1053 <https://github.com/Toni-cs/ecg-calibration-transportability>.
1054 The protocol, both amendments (A1, A2), and a per-file archival-status re-
1055 port are publicly archived at <https://doi.org/10.17605/OSF.IO/53H64>.
1056 docs/osf_archive_manifest.json is the internal snapshot taken on 2026-09-
1057 05; it is not a substitute for the public record.

1058 **Code availability**

1059 The full pipeline code, preprocessing (reproducible from the public source
1060 databases), patient-level split generation, model training, the calibration-method
1061 implementations, and all statistical analyses, together with the environment
1062 lock file, the patient-level split indices for all five dataset variants and five
1063 random seeds, and selected per-experiment result artifacts, is publicly available
1064 at <https://github.com/Toni-cs/ecg-calibration-transportability>.
1065 Trained model checkpoints for the 60-experiment transfer grid are provided as
1066 release assets of the same repository. The repository will be archived on Zenodo
1067 with a DOI upon acceptance. No new raw waveforms were collected or generated;

1068 the patient-level split indices and label-mapping tables released with the code
1069 fully determine the data configuration used in every experiment.

1070 **Declaration of generative AI and AI-assisted technologies in the writing pro-**
1071 **cess**

1072 All scientific work reported in this study—the study design, the pre-registered
1073 protocol and its amendments, the implementation and execution of the 60-
1074 experiment transfer grid, the statistical analysis, and the interpretation of the
1075 results—was performed by the author. A large language model assistant (GLM)
1076 was used in two supervised roles, neither of which produced scientific content:
1077 as a coding assistant that drafted and refactored parts of the analysis and plotting
1078 code (Section 4.10), and as a language-editing aid that improved the wording and
1079 readability of text the author had written. The tool generated no experimental data,
1080 results, or conclusions, and made no analytical or interpretative decisions; every
1081 reported number is produced by the author’s scripts and is reproducible from the
1082 public repository. After using the tool, the author reviewed and edited all affected
1083 code and text as needed and takes full responsibility for the content of the pub-
1084 lished article.

1085 **CRedit authorship contribution statement**

1086 **Toni Guan:** Conceptualization, Methodology, Software, Validation, Formal
1087 analysis, Investigation, Data curation, Writing – original draft, Writing – review
1088 & editing, Visualization, Project administration.

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1096 **Declaration of competing interest**

1097 The author declares no competing financial or personal interests.

1098 The 60 per-seed reliability diagrams (6 transfer pairs $\times$ 2 architectures $\times$ 5
1099 seeds; calibration curves, reliability diagrams, and Δ ECE bars per experiment)
1100 are provided as a separate supplementary file (supplementary.pdf, §1) and are
1101 additionally released with the code repository.

1102 **References**

- 1103 [1] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. Dillon,
1104 B. Lakshminarayanan, J. Snoek, “Can you trust your model’s uncertainty?
1105 Evaluating predictive uncertainty under dataset shift,” *NeurIPS*, 2019.
1106 arXiv:1906.02530.
- 1107 [2] C. Guo, G. Pleiss, Y. Sun, K. Weinberger, “On calibration of modern neural
1108 networks,” *ICML*, PMLR 70:1321–1330, 2017. arXiv:1706.04599.

- 1109 [3] P. Wagner, N. Strodthoff, R.-D. Bousseljot, et al., “PTB-XL, a large
1110 publicly available electrocardiography dataset,” *Scientific Data*, 7:154,
1111 2020. doi:10.1038/s41597-020-0495-6.
- 1112 [4] J. Zheng, J. Guo, H. Chu, “A large scale 12-lead electrocardiogram
1113 database for arrhythmia study (Chapman-Shaoxing e-cardiology),” version
1114 1.0.0, *PhysioNet*, 2022. doi:10.13026/wgex-er52.
- 1115 [5] F. Liu, C. Liu, L. Zhao, et al., “An open access database for evaluating the
1116 algorithms of electrocardiogram rhythm and morphology abnormality
1117 detection,” *Journal of Medical Imaging and Health Informatics*,
1118 8(7):1368–1373, 2018. doi:10.1166/jmihi.2018.2442.
- 1119 [6] E. A. P. Alday, A. Gu, A. J. Shah, et al., “Classification of 12-lead ECGs:
1120 the PhysioNet/Computing in Cardiology Challenge 2020,” *Physiological
1121 Measurement*, 41(12):124003, 2020. doi:10.1088/1361-6579/abc960.
- 1122 [7] A. L. Goldberger et al., “PhysioBank, PhysioToolkit, and PhysioNet:
1123 components of a new research resource for complex physiologic signals,”
1124 *Circulation*, 101(23):e215–e220, 2000.
- 1125 [8] J. Nixon, M. W. Dusenberry, L. Zhang, et al., “Measuring calibration in
1126 deep learning,” *CVPR Workshops*, pp. 38–41, 2019. arXiv:1904.01685.
- 1127 [9] M. Kull, M. Perello-Nieto, M. Kangse, T. S. Filho, P. Song, P. Flach,
1128 “Beyond temperature scaling: obtaining well-calibrated multi-class
1129 probabilities with Dirichlet calibration,” *NeurIPS*, 2019. arXiv:1910.12656.

- 1130 [10] A. Alexandari, A. Kundaje, A. Shrikumar, “Maximum likelihood with
1131 bias-corrected calibration is hard-to-beat at label shift adaptation,” *ICML*,
1132 PMLR 119:222–232, 2020. arXiv:1901.06852.
- 1133 [11] Z. Lipton, Y.-X. Wang, A. Smola, “Detecting and correcting for label shift
1134 with black box predictors,” *ICML*, PMLR 80:1944–1953, 2018.
1135 arXiv:1802.03916.
- 1136 [12] M. Barandas, L. Famiglini, A. Campagner, D. Folgado, R. Simão,
1137 F. Cabitza, H. Gamboa, “Evaluation of uncertainty quantification methods
1138 in multi-label classification: a case study with automatic diagnosis of
1139 electrocardiogram,” *Information Fusion*, 101:101978, 2024.
1140 doi:10.1016/j.inffus.2023.101978.
- 1141 [13] J. F. Vranken et al., “Uncertainty estimation for deep learning-based
1142 automated analysis of 12-lead electrocardiograms,” *European Heart*
1143 *Journal – Digital Health*, 2(3):423–434, 2021. doi:10.1093/ehjdh/ztab045.
- 1144 [14] M. Reyna, N. Strodthoff, P. Wagner, et al., “Issues in the automated
1145 classification of multilead ECGs using heterogeneous labels and
1146 populations,” *Physiological Measurement*, 43(8):084001, 2022.
1147 doi:10.1088/1361-6579/ac79fd.
- 1148 [15] B. Efron, “Better bootstrap confidence intervals,” *Journal of the American*
1149 *Statistical Association*, 82(397):171–185, 1987.
1150 doi:10.1080/01621459.1987.10478410.
- 1151 [16] B. Van Calster, D. J. Steyerberg, S. Collins, R. D. Riley, E. W. Steyer, “A
1152 calibration hierarchy for risk models was defined: from utopia to empirical

- 1153 data,” *Journal of Clinical Epidemiology*, 74:167–176, 2016.
1154 doi:10.1016/j.jclinepi.2015.12.005.
- 1155 [17] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, G. S. Collins,
1156 “Calibration: the Achilles heel of predictive analytics,” *BMC Medicine*,
1157 17:230, 2019. doi:10.1186/s12916-019-1466-7.
- 1158 [18] A. Kumar, P. Liang, T. Ma, “Verified uncertainty calibration,” *NeurIPS*,
1159 2019. arXiv:1909.10155.
- 1160 [19] N. Strodthoff, P. Wagner, T. Schaeffter, W. Samek, “Deep learning for ECG
1161 analysis: benchmarks and insights from PTB-XL,” *IEEE Journal of*
1162 *Biomedical and Health Informatics*, 25(5):1519–1528, 2021.
1163 doi:10.1109/JBHI.2020.3022989.
- 1164 [20] M. Haekal, S. N. Khotimah, G. R. F. Suwandi, F. Haryanto,
1165 “RR-interval-based atrial fibrillation detection and burden estimation:
1166 cross-dataset validation and calibration-aware probability analysis,”
1167 *Physiological Measurement*, 2026. doi:10.1088/1361-6579/ae99aa.
- 1168 [21] K. Patel, P. Beedala, “Calibration drift under cross-institutional deployment:
1169 an external validation framework for ICU mortality prediction across
1170 MIMIC-IV and eICU,” *medRxiv* (preprint), 2026.
1171 doi:10.64898/2026.05.03.26352335.
- 1172 [22] X. Wang, M. Long, J. Wang, M. I. Jordan, “TransCal: transferability
1173 estimation for auto-selecting transfer calibration,” *NeurIPS*, 2020.
1174 arXiv:2007.08259.

- 1175 [23] S. Garg, Y. Wu, S. Balakrishnan, Z. Lipton, “A unified view of label shift
1176 estimation,” *NeurIPS*, 2020. arXiv:2003.07554.
- 1177 [24] T. Popordanoska, G. Radevski, T. Tuytelaars, et al., “LaSCal: label-shift
1178 calibration for uncertainty quantification,” *NeurIPS*, 2024.
- 1179 [25] J. T. Moreno-Torres, T. Raeder, R. Alaiz-Rodriguez, N. Chawla, F. Herrera,
1180 “A unifying view on dataset shift in classification,” *Pattern Recognition*,
1181 45(1):521–530, 2012. doi:10.1016/j.patcog.2011.05.019.
- 1182 [26] [Pre-registered protocol, publicly archived] “ECG calibration boundary
1183 study – experiment protocol,” 2026. Hash manifest:
1184 docs/osf_archive_manifest.json (local snapshot; archived at
1185 <https://doi.org/10.17605/OSF.IO/53H64>); revision A1 full text in
1186 docs/PROTOCOL_AMENDMENT_A1_PREREG_RELOCATION.md.
- 1187 [27] B. Nosek, C. Ebersole, A. DeHaven, D. Mellor, “The preregistration
1188 revolution,” *PNAS*, 115(11):2600–2606, 2018.
1189 doi:10.1073/pnas.1708274114.
- 1190 [28] D. Lakens, “The practical alternative to the p value is the correctly used p
1191 value,” *Perspectives on Psychological Science*, 16(3):639–648, 2021.
1192 doi:10.1177/1745691620958012.
- 1193 [29] J. Li, A. D. Aguirre, V. M. Junior, et al., “An Electrocardiogram Foundation
1194 Model Built on over 10 Million Recordings,” *NEJM AI*, 2(7):AIoa2401033,
1195 2025. doi:10.1056/AIoa2401033.
- 1196 [30] J. C. McHugh, B. Wang, et al., “ECG-FM: an open electrocardiogram
1197 foundation model,” *arXiv preprint* 2408.05178, 2024.

- 1198 [31] J. Jin, et al., “Reading your heart: learning ECG words and sentences via
1199 pre-training ECG language model (HeartLang),” *ICLR*, 2025.
1200 arXiv:2502.10707.
- 1201 [32] C. V. Nguyen, C. D. Do, “Transfer learning in ECG diagnosis: is it
1202 effective?” *PLOS ONE*, 20(5):e0316043, 2025.
1203 doi:10.1371/journal.pone.0316043.
- 1204 [33] S. Vavaroutas, et al., “ADAPTOOD: uncertainty-aware fine-tuning for
1205 out-of-distribution ECG time series models,” *arXiv preprint* 2606.04164,
1206 2026.
- 1207 [34] W. Jiang, et al., “Test-time adaptation for ECG classification via SQI-gated
1208 self-training and beat-rhythm consistency,” *arXiv preprint* 2608.23347,
1209 2026 (accepted at MICCAI 2026).
- 1210 [35] N. Nemati, “Comparative analysis of data augmentation for clinical ECG
1211 classification with STAR (Sinusoidal Time–Amplitude Resampling),” *arXiv*
1212 *preprint* 2510.24740, 2025.
- 1213 [36] X. Liu, et al., “PEACE: knowledge-guided cross-modal fusion for
1214 adult-to-pediatric ECG transfer via label-conditioned contrastive
1215 alignment,” *arXiv preprint* 2605.00647, 2026.
- 1216 [37] J. Song, J.-H. Jang, B. T. Lee, D. Hong, J. Kwon, Y.-Y. Jo, “Foundation
1217 models for electrocardiograms,” *arXiv preprint* 2407.07110, 2024.